



KALDRIVON

Kaldrивon MiniSMO

User Guide

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1. Introduction

This guide covers Kaldrivon MiniSMO version 0.4.0 for Windows 11. It explains how to install the application, create the built-in Demo Network, connect real NETCONF-managed network functions, read their configuration and state, and work with the simulated fault, performance and scenario features.

The guide is written for lab engineers, developers, testers, students and solution architects who need a small, local management environment for O-RAN-style network functions. You don't need to be a NETCONF expert; the NETCONF chapters start from the basics.

Conventions used in this guide:

Convention	Meaning
CREATE DEMO NETWORK	A button or menu item in the application
<code>%LOCALAPPDATA%</code>	A Windows path or a value you type
<code>SYSTEM > License</code>	Navigation: the <i>License</i> page in the <i>SYSTEM</i> group

Synthetic data

Everything the Demo Network produces (configuration, alarms, counters, software versions) is synthetic. It is realistic enough to learn and test with, but it does not come from real equipment.

2. What Kaldrivon MiniSMO Is

Kaldrivon MiniSMO is a lightweight, O-RAN-oriented management, simulation and test environment that runs on a single Windows PC. It has two halves that you can use together or separately:

- **A real NETCONF manager.** MiniSMO connects to external network functions over NETCONF/SSH (normally TCP port 830), performs the hello exchange, discovers capabilities, and runs `<get>` and `<get-config>`. Every message is recorded in the NETCONF Log with its full XML.
- **A built-in simulator.** The Demo Network is a small simulated RAN (O-CU-CP, O-CU-UP, two O-DUs and ten O-RUs) with editable configuration, alarms, smoothly changing performance counters and scripted scenarios such as cell outage and congestion.

Typical uses:

- Checking that a lab O-RU or O-DU answers NETCONF correctly before integrating it with a larger SMO.
- Looking at the capabilities and YANG modules a device advertises.
- Teaching or learning SMO concepts (FM, PM, CM, topology) without access to live equipment.
- Demonstrating operational workflows with repeatable scenarios.

MiniSMO is not a production network management system and does not replace an operator SMO. See [Known Limitations](#).

3. Version 0.4 Scope

Version 0.4.0 includes:

Area	What is included
External NFs	Add, edit and delete NETCONF-managed NFs; staged connection test; connect/disconnect; capability discovery; <code><get></code> ; <code><get-config></code> on advertised datastores
NETCONF	NETCONF over SSH, base 1.0 and 1.1 framing, password, keyboard-interactive and SSH key authentication, host-key pinning on first connect, YANG library discovery (RFC 8525), NETCONF Call Home (RFC 8071)
External monitoring	FM, CM and PM from external NFs: NETCONF notifications (RFC 5277/6470), YANG-Push (RFC 8639/8641), O-RAN FM/PM, ietf-alarms, 3GPP AlarmList, PM pull from the NF's model, with reconnect, replay, gap tracking and freshness
VES collector	VES 7.2.1 event listener (fault, perf3gpp, heartbeat, pnfRegistration, 3GPP stndDefined), attribution by sourceName, and the NF's VES destination set when the NF is added (can be turned off) with the standard models only: 3GPP <code>NtfSubscriptionControl</code> and, for O-RUs, the O-RAN WG4 configured subscription
Demo Network	14 simulated NFs with topology, configuration, alarms, PM and scenarios
Configuration	GET / VALIDATE & STAGE / APPLY / REVERT for simulated NFs; parameter form and XML for external NFs, editable once enabled per NF
Fault Management	One Alarm Console for simulated, external and internal alarms; external alarms from NETCONF notifications, alarm-list reads and VES; alarm resync; internal supervision and health alarms; generate and clear simulated alarms
Performance Management	Nine simulated counters with 1, 5, 15 or 30 second intervals; external PM from YANG-Push, the NF's own PM model (including O-RAN WG4 O-RU measurements), O-RAN PM notifications, VES perf3gpp and 3GPP measurement files
Scenarios	Six built-in scenarios with RUN, PAUSE, RESUME, STOP and RESET
Logs	NETCONF Log with search, filter, copy, export and clear; Audit Log
YANG	YANG model registry (<i>SYSTEM > Settings > YANG Models</i>)
Editions	Kaldrivon Mini SMO Community : free, up to 5 external NFs, all essential FM, PM and CM. Kaldrivon Mini SMO Pro : any valid signed <code>.kLicense</code> ; more NFs plus the operations features below

Area	What is included
Operations (Community)	Inventory tree, NF identity and capabilities, global search, alarm CSV export, manual configuration backups and single-NF compare, audit log, in-app notifications, system health, network health report, purge of FM, PM, CM or log data. A small daily allowance of every Pro feature
Operations (Pro)	Northbound REST API, O-RAN R1 for rApps (SME and DME), webhooks, automation rules and schedules, approvals, alarm correlation and flapping detection, KPI rankings, advanced compare, desired state and drift, templates, bulk operations, rollback, SW campaign planning, scheduled reports, Investigate, diagnostics, advanced retention and audit export
Capacity & Storage	Capacity test of this PC, storage usage by category, storage health, growth forecast, tiered PM retention (raw, 5-minute, hourly), automatic and manual cleanup, capacity and storage reports
Help	This offline Help Center, the User Guide (HTML and PDF) and a Quick Start guide

Version 0.4.0 changes external devices only in two ways: the VES collector address, set automatically when you add the NF, and configuration changes you enable and confirm per NF. One more request acts on the NF: a file upload of a measurement file (`file-upload`), sent only after you confirm it under **FILES ON NF...**; it changes no configuration. Pro templates, remediation, rollback and automation actions all go through the same per-NF enablement, preview and confirmation. See [How MiniSMO Writes to External NFs](#).

4. System Requirements

Item	Requirement
Operating system	Windows 11 x64 (22H2 or later recommended)
Processor	Any x64 CPU with 2 or more cores
Memory	4 GB RAM minimum, 8 GB recommended
Disk	About 300 MB for the application, plus room for logs and PM history
Display	1366 × 768 or larger
Components	Microsoft Edge WebView2 Runtime (included with Windows 11)
Network	Only needed to reach external NFs. Help and the Demo Network work offline
Permissions	Administrator rights to install; a standard user account to run

You do not need to install Python, .NET or any other runtime. The installer contains everything MiniSMO needs.

To reach an external NF, the PC must be able to open a TCP connection to the device's NETCONF port (830 by default). Corporate firewalls and VPN split-tunnel rules often block this. See [NETCONF Connection Requirements](#).

5. Installation

1. Download `Kaldrivon-MiniSMO-0.4.0-Setup.exe`.
2. Optional but recommended: verify the download. The release includes `Kaldrivon-MiniSMO-0.4.0-Setup.exe.sha256`. In PowerShell:

```
Get-FileHash .\Kaldrivon-MiniSMO-0.4.0-Setup.exe -Algorithm SHA256
```

The hash must match the value in the `.sha256` file.
3. Run the installer. Windows may ask for administrator permission.
4. Accept the license agreement, keep the default folder `C:\Program Files\Kaldrivon\MiniSMO` or choose another, and choose whether you want a desktop shortcut.
5. Select **Install**, then **Finish**. You can launch MiniSMO directly from the last page of the installer.

The installer creates a Start Menu entry **Kaldrivon MiniSMO**, an optional desktop shortcut, and an entry in *Settings > Apps > Installed apps* for uninstalling.

SmartScreen

If the release you received is not code-signed, Windows SmartScreen may show "Windows protected your PC". Select **More info** and check that the file name and hash match what Kaldrivon provided before choosing **Run anyway**.

6. First Launch

When you start MiniSMO:

1. The application checks whether its local backend service is running on `http://127.0.0.1:8765`.
2. If not, it starts the bundled backend (`KaldrivonMiniSMO.Backend.exe`) in the background. No console window appears.
3. On first run the backend creates its data folders under `%LOCALAPPDATA%\Kaldrivon\MiniSMO`, creates the database and applies migrations.
4. The Dashboard opens.

On a fresh installation the Dashboard shows:

- `Community` | `Managed NFs 0/5` under the page title and in the title bar.
- The **LICENSE AND CAPACITY** card with Edition *Community* and Managed NFs *0/5*.
- The **ADD EXTERNAL NF** button. The Demo Network is created under *SIMULATION > Demo Network*.

If the backend does not start within about 30 seconds, MiniSMO shows a diagnostic panel with the reason and an **Open startup troubleshooting** link. See [Backend Startup Problems](#).

Screenshot placeholder: **Dashboard on first launch: Community, Managed NFs 0/5**

To be captured from the Windows 0.4.0 build (docs/images/screenshots/dashboard-first-run.png)
before public release.

7. Application Layout

The main window has a navigation pane on the left and the working area on the right.

Group	Pages (tabs)
OVERVIEW	Dashboard
NETWORK	Network Functions (per NF: Overview, Capabilities, Monitoring); Topology & Inventory (Topology, Inventory)
FCAPS	Alarms; Configuration (Configuration, Configuration Operations); Performance
TOOLS	Automation, Analytics, Software Campaigns, API & Webhooks (all marked PRO)
LOG	NETCONF Log, Audit Log
SIMULATION	Demo Network, Scenarios
SYSTEM	License; Health & Capacity (Capacity & Storage, Reports); Settings (General, Storage & Retention, VES Collector, YANG Models)

Help and **About** are at the bottom of the navigation pane. The search box at the top of the pane finds NFs, alarms and cells. Pages with several sections show them as tabs across the top.

The title bar shows the edition chip (COMMUNITY or PRO), the backend state, the license indicator with the effective edition and external NF usage, for example **Community** | Managed NFs 2/5, and the notifications bell.

Confirmations and exports. Buttons that run something (exports, diagnostics, reports, snapshots, running a schedule, testing a webhook, analyses) ask for confirmation first. After an export a dialog shows where the file was saved, with **Open file** and **Show in folder**.

Most pages have a **?** help button in their header. It opens the help topic for that page. **F1** does the same from anywhere on the page, including the Add External NF dialog.

Keyboard basics:

Key	Action
F1	Help for the current page
Ctrl+F	Focus the search box (Help, NETCONF Log, Alarm Console, Network Functions capabilities)
F5	Refresh the current page
Esc	Close a dialog

8. Dashboard

OVERVIEW > Dashboard (page title **Network Overview**) gives a one-screen summary of everything MiniSMO manages. It refreshes every 5 seconds. The line under the title is the license indicator, for example **Community | Managed NFs 3/5** or **Pro | Managed NFs 12/25**. If a license has expired or is not valid, it says so, for example **Community (license expired) | Managed NFs 3/5**.

Summary tiles

Tile	Meaning
Network functions	All NFs, simulated and external
Simulated	Demo Network NFs (synthetic data)
External	Real NETCONF NFs you added. These count against your license
Active alarms	Alarms that have not been cleared, all severities
Critical / Major	Active alarms at that severity

Cards

Card	Shows
ACTIVE ALARMS	The active alarms among the eight most recent ones. Select an alarm or OPEN ALARM CONSOLE to open <i>FCAPS > Alarms</i>
NETWORK FUNCTION STATUS	Every NF with its type, origin (simulated or external) and connection state
RECENT NETCONF ACTIVITY	The last eight NETCONF log entries. Select one to open it in the NETCONF Log
WHAT NEEDS ATTENTION · SYSTEM	Items that need a look now (for example disconnected NFs, critical alarms, failed jobs, storage state) and the System health state (HEALTHY, DEGRADED or CRITICAL). See Reports and System Health
LICENSE AND CAPACITY	Edition, Managed NFs (for example 3 / 5), Technical capacity from the latest capacity test (for example ~180 external, or Not tested yet) and Storage (Healthy, Warning, Critical or Cleanup running). LICENSE opens <i>SYSTEM > License</i> ; HEALTH & CAPACITY opens <i>SYSTEM > Health & Capacity</i>
EXTERNAL NF MONITORING	How many external NFs are monitored, and how many are connected, receiving, subscribed (quiet), stale or reconnecting, unsupported or polling. NETWORK FUNCTIONS > MONITORING opens the Monitoring tab of Network Functions
DATA ORIGIN	Which data is synthetic (Demo Network) and which comes from external NFs

See [Capacity & Storage Overview](#).

Actions

- **ADD EXTERNAL NF** (top right): opens the Add External NF dialog.
- The Demo Network is created and managed under *SIMULATION > Demo Network*.

Screenshot placeholder: **Dashboard with the Demo Network running**

To be captured from the Windows 0.4.0 build (docs/images/screenshots/dashboard-demo.png) before public release.

9. Creating the Demo Network

The Demo Network is a small simulated O-RAN deployment built into MiniSMO. It lets you explore every page of the application without any real equipment.

To create it, open *SIMULATION > Demo Network* and select **CREATE DEMO NETWORK**. MiniSMO creates 14 simulated network functions:

Type	Count	Names
O-CU-CP	1	O-CU-CP-001
O-CU-UP	1	O-CU-UP-001
O-DU	2	O-DU-001, O-DU-002
O-RU	10	O-RU-001 to O-RU-010

Every simulated NF starts **UNLOCKED**, **ENABLED** and **CONNECTED**, carries the **SIMULATED** badge, and immediately starts producing performance data.

The Demo Network does **not** use external NF license capacity. After creating it, the license indicator still shows **Managed NFs 0/5**.

Managing the Demo Network (*SIMULATION > Demo Network*):

- **CREATE DEMO NETWORK** creates the 14 NFs, or **RESTORE n MISSING** recreates only the simulated NFs you deleted.
- **Alarm activity** sets how often simulated NFs raise and clear alarms on their own (off, low, normal, high); it is saved immediately.
- **RESET** deletes all 14 simulated NFs with their configuration, alarms and PM history, and creates them again in their initial state. You are asked to confirm.
- **DELETE** removes the simulated NFs without recreating them. You are asked to confirm.

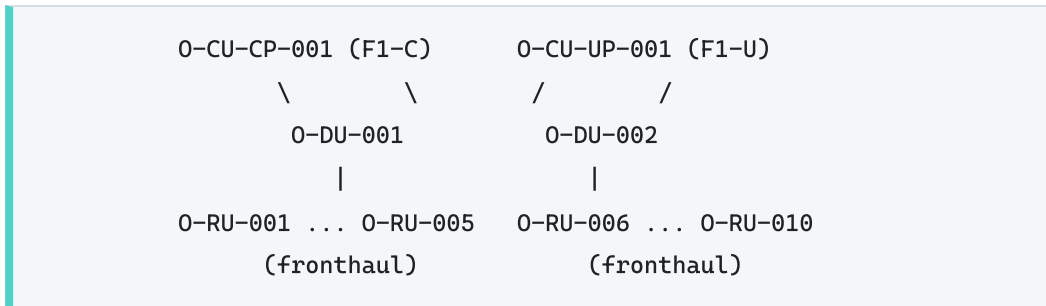
External NFs are never affected by these actions.

Note

The Demo Network names (O-RU-001 and so on) are reserved while the Demo Network exists. If you already have an external NF with one of those names, MiniSMO refuses to create the Demo Network and tells you which name clashes.

10. Understanding the Demo Topology

The Demo Network is wired like a typical small split-7.2 deployment:



- **O-DU-001** serves O-RU-001 through O-RU-005.
- **O-DU-002** serves O-RU-006 through O-RU-010.
- Both O-DUs connect to **O-CU-CP-001** (control plane, F1-C) and **O-CU-UP-001** (user plane, F1-U).

Scenarios use this structure. For example CELL OUTAGE takes down O-RU-007 under O-DU-002, and HIGH CPU LOAD stresses O-DU-001. See [Running Scenarios](#) and [Topology View](#).

11. Network Function Inventory

NETWORK > Network Functions lists every NF MiniSMO knows about, simulated and external, in one table.

Column	Meaning
NAME / ADDRESS	Unique name you gave the NF (or the Demo Network name), and host and port for external NFs
TYPE	O-RU, O-DU, O-CU-CP, O-CU-UP or GENERIC NETCONF NF
ORIGIN	Simulated (Demo Network) or external (real device over NETCONF)
CONNECTION	CONNECTED or DISCONNECTED
MONITORING	External NFs only: ONGOING, CONNECTING, STARTING, RECONNECTING, FAILED, STOPPED or NOT MONITORED. Empty for simulated NFs

Use the **Show** box to list All, Simulated or External NFs.

Select an NF to open its detail pane, with three tabs:

- **Overview:** identity, state, last error, NETCONF session ID and version, host key fingerprint, and **FORGET HOST KEY**.
- **Capabilities:** the NETCONF capabilities the device advertised in its last hello. See [NETCONF Capability Discovery](#).
- **Monitoring:** FM, CM and PM monitoring, VES and file-based PM of an external NF. See [Monitoring External NFs](#).

NETCONF reads (**GET** and **GET-CONFIG** with a subtree filter) are on the **NETCONF read** tab of *FCAPS > Configuration*.

Toolbar actions for the selected NF (the ones that do not fit move to ...):

Action	What it does
Connect / Disconnect / Test connection	External NFs only
Edit	Change an external NF's settings
Delete	Asks for confirmation, with the option to stop the NF sending VES to MiniSMO (see Pointing an NF at the Collector). A progress window then shows each step: VES destination, monitoring stopped, history deleted (row count), NF removed. It closes only when the deletion has finished; a second delete of the same NF is refused while one runs. Deleting an external NF frees a license slot
Resync alarms	Reads the NF's active alarm list again and reconciles the alarm view. See Monitoring External NFs

Action	What it does
Discover identity	Reads vendor, model, serial number and versions again. See NF Identity and Cells
Site / region	Sets the NF's site and region for the network tree
Capabilities	Vendor-neutral summary of what the NF can technically do (this is not the Capabilities tab)
Back up config	Stores a configuration backup. See Configuration Backups
Investigate (Pro)	See Investigate
Diagnostics (Pro)	Asks for confirmation, then writes a diagnostic ZIP. See Diagnostic Collection
Export CSV	Asks for confirmation, then writes the NF inventory to a CSV file and shows Open file / Show in folder

12. Adding an External NF

12.1 Adding an External NF

An *external NF* is a real network function that MiniSMO manages over NETCONF/SSH. To add one:

1. Select **ADD EXTERNAL NF** on the Dashboard or on *NETWORK > Network Functions*.
2. Fill in the form:

Field	What to enter
Name	A unique label, for example <code>LAB-0-RU-01</code>
NF Type	O-RU, O-DU, O-CU-CP, O-CU-UP or GENERIC NETCONF NF
Hostname/IP	DNS name or IP address of the device's management interface
Authentication	Password, Keyboard-interactive or SSH key. For SSH key, LOAD KEY FILE... and enter the key passphrase if the key is encrypted. See SSH Authentication Methods
NETCONF Port	Usually 830 (the default is taken from Settings)
Username	NETCONF/SSH user on the device
Password	That user's password (not used with SSH key). Stored encrypted with Windows DPAPI, never in plain text
Description	Optional free text

3. Select **TEST CONNECTION**. MiniSMO runs every connection stage and shows which ones passed. See [NETCONF Connection Test](#).
4. Select **SAVE**. MiniSMO checks your external NF capacity before saving. If you are at the limit, you see *Community supports up to 5 external managed NFs (5/5 in use). Upgrade to Kaldrivon Mini SMO Pro to manage additional NFs.* (with Pro, the licensed limit) and nothing is saved.
5. MiniSMO integrates the NF straight away: it connects, starts FM / CM / PM monitoring and points the NF's VES destination at the collector. A dialog shows each step. If the device is unreachable, the NF is saved anyway and shows **DISCONNECTED**; choose **CONNECT** later. Turn automatic integration off in *SYSTEM > Settings > General*, section **Integration**.

The **Help with NETCONF connection** link in the dialog opens [NETCONF Connection Requirements](#). **F1** opens this topic.

Saving does not require a successful test, so you can add devices that are currently powered off. They still use a license slot.

12.2 Example: Adding an External O-RU

This walkthrough adds a lab O-RU and reads its configuration.

1. Enter the details

Field	Value
Name	LAB-O-RU-01
NF Type	O-RU
Hostname/IP	192.168.10.50
NETCONF Port	830
Username	admin
Password	*****

2. TEST CONNECTION. The result list shows each stage. A healthy device looks like this:

Stage	Result	Example detail
Host Resolution	✓	192.168.10.50 -> 192.168.10.50
TCP Connection	✓	connected to 192.168.10.50:830
SSH	✓	SSH-2.0-OpenSSH_8.x; host key ssh-ed25519 SHA256:...
Authentication	✓	authenticated as 'admin'
NETCONF subsystem	✓	netconf subsystem opened
Server Hello	✓	session-id 42; NETCONF 1.1; 37 capabilities

Below the list MiniSMO shows the negotiated **NETCONF version** and the **capability count**.

If a stage fails, the stages after it show *not reached*. What to check at each stage:

Failed stage	Typical message	First things to check
Host Resolution	cannot resolve 'lab-ru-01'	Spelling; use the IP address instead; DNS/VPN
TCP Connection	timed out connecting to 192.168.10.50:830	Device powered and reachable (ping), port number, firewalls between you and the device
SSH	SSH handshake failed	Something other than SSH listens on that port; very old or restricted SSH algorithms on the device
Authentication	authentication failed (wrong username or password)	Credentials; account locked; device allows only public-key login

Failed stage	Typical message	First things to check
NETCONF subsystem	the 'netconf' subsystem was refused	SSH works but NETCONF is not enabled for that port or user
Server Hello	timed out / not well-formed XML	Device NETCONF service unhealthy or non-standard; try again after restarting it

3. SAVE. The O-RU appears in *Network Functions* as an external NF and the license indicator goes up by one. With automatic integration on (the default), MiniSMO connects, starts monitoring and points the NF's VES destination at the collector; a dialog shows each step.

4. CONNECT (only if integration is off or the device was unreachable). MiniSMO opens a session, stores the capabilities and pins the device's SSH host key. The row changes to CONNECTED.

5. GET. In *FCAPS > Configuration*, select the NF, open the **NETCONF read** tab and select **GET**. MiniSMO sends `<get/>` and shows the operational and configuration data returned.

6. GET-CONFIG. Choose the **running** datastore and select **GET-CONFIG**. Only datastores the device advertised are selectable. See [Running GET-CONFIG](#).

All requests and replies are in *LOG > NETCONF Log*.

12.3 NETCONF Call Home

Normally MiniSMO opens the connection to the NF. With *NETCONF Call Home* (RFC 8071) the NF opens the TCP connection to MiniSMO instead, which suits NFs behind NAT or a firewall that blocks incoming connections. MiniSMO stays the SSH client and the NETCONF client: after the connection arrives, everything is the same as for any external NF. It is the same NF record, with the same credentials, license slot, capability discovery, FM / CM / PM monitoring, VES, configuration history and PM discovery.

Only NFs you registered are accepted. Call Home is off until you switch it on in *SYSTEM > Settings*, section **NETCONF Call Home: Listen for Call Home connections**, the listen address, the port (default 4334, IANA `netconf-ch-ssh`) and how long a connect waits for the NF to call. The installer opens TCP 4334 in Windows Firewall for the backend (Domain and Private networks) when **Allow NFs to call MiniSMO** is ticked during setup; it is removed on uninstall. For another port or network, the section shows the PowerShell command. A caller that is not one of your NFs is closed during the SSH key exchange, before MiniSMO sends any credential. It is listed under **REFUSED CALLERS** (address, SSH host key fingerprint, reason). After comparing the fingerprint with the one the NF itself shows, **ADD AS NF...** opens *Add External NF* in Call Home mode with the fingerprint and address filled in.

To add one, choose **Connection: The NF calls MiniSMO (NETCONF Call Home, RFC 8071)** in *Add External NF*. The port field disappears, the hostname becomes

optional, and the **CALL HOME IDENTITY (ALLOW-LIST)** fields appear. The NF must say how its caller is recognised:

Field	Meaning
SSH host key fingerprint	SHA256: . . . as the NF shows it. Checked during the key exchange; the strongest check. A different key is never accepted
Serial number, hardware id	Checked after login against what the NF reports about itself (<code>ietf-hardware serial-num</code> and <code>hardware-rev</code>). If the NF does not report them, it is refused. Without a fingerprint these need pin the host key on the first call , so the first verified call fixes the key for good
Allowed source	Optional IP address or network (for example <code>10.20.0.0/24</code>) the calls must come from

Hostname and port are not needed: the address is learned from each call. One session per NF is used for monitoring and for reads. If the NF disconnects it calls again and monitoring resumes. An NF that calls with the wrong identity is refused and listed as refused; its host key is not pinned.

EDIT on the NF changes the connection mode and these identity fields later; changing them makes the NF prove its identity again on its next call. The NF's details show the connection mode, its identity and the last call (time and address). **TEST CONNECTION** does not apply to a Call Home NF: it is checked when it calls, and the integration dialog after **SAVE** waits for that call.

The same is available through the API: `POST /api/nfs/external` with `connect_mode: "call-home"`, `host_key_fingerprint`, `ch_serial`, `ch_hardware_id`, `ch_source`, `ch_trust_first_use`; the listener state is `GET /api/callhome`.

12.4 SSH Authentication Methods

When you add an external NF you choose how MiniSMO signs in over SSH:

Method	Use it when
Password	The usual case. If the NF only offers keyboard-interactive, MiniSMO falls back to it with the same password
Keyboard-interactive	The NF asks for the password through a keyboard-interactive prompt, possibly with non-standard prompt text (O-RAN-SC NTS does this). Every prompt is answered with the password
SSH key	The NF has your public key in its authorized keys. Load the private key file (OpenSSH or PEM; Ed25519, ECDSA or RSA) and enter its passphrase if it is encrypted

Passwords, private keys and passphrases are stored protected for your Windows user (DPAPI) and never shown again. They are masked in all logs. Nothing about an NF

(address, port, username, secrets) is built into MiniSMO: it is all entered when you add the NF.

13. NETCONF Connection Requirements

13.1 NETCONF Connection Requirements

For MiniSMO to manage a device, all of the following must be true:

1. **Network reachability.** Your PC can reach the device's management IP address, and nothing between them (Windows Firewall, corporate firewall, VPN, security groups) blocks the NETCONF port.
2. **NETCONF over SSH is enabled** on the device, normally on TCP port **830**. Some devices use port 22 with the `netconf` SSH subsystem instead.
3. **A NETCONF user account** exists, with a password or an SSH key. MiniSMO supports password, keyboard-interactive and public-key authentication (see [SSH Authentication Methods](#)).
4. **The device speaks standard NETCONF:** it sends a `<hello>` with at least the base 1.0 or base 1.1 capability.
5. **Read permission** for `<get>` and `<get-config>`. NACM on the device may limit what the account can see.

Quick checks from a Windows PowerShell prompt:

```
Test-NetConnection 192.168.10.50 -Port 830
ssh -p 830 admin@192.168.10.50 -s netconf
```

The first should report `TcpTestSucceeded : True`. The second should print the device's XML `<hello>` after login (press Ctrl+C to exit).

13.2 What is NETCONF?

NETCONF (Network Configuration Protocol, RFC 6241) is the IETF protocol for reading and changing the configuration and state of network devices. O-RAN uses NETCONF with YANG data models for O-RU management (the M-plane) and on the O1 interface between the SMO and O-RAN network functions.

The basics:

- The client (here, MiniSMO) and the server (the device) exchange XML messages.
- The client sends **RPCs** such as `<get>` or `<get-config>`, each with a `message-id`. The server answers with an `<rpc-reply>` carrying the same `message-id`, containing either data, `<ok/>` or an `<rpc-error>`.
- The structure of the data is defined by **YANG** modules. The device tells the client which modules it supports in its capabilities.
- Configuration is kept in **datastores**: *running* always exists; *candidate* and *startup* are optional.

13.3 Why Port 830?

TCP port **830** is the port IANA assigned to NETCONF over SSH (RFC 6242). Most devices listen for NETCONF there, which is why MiniSMO uses 830 as the default.

Some devices instead offer NETCONF on the normal SSH port 22 through the `netconf` SSH subsystem, or on a vendor-chosen port. If your device uses a different port, enter it in the **NETCONF Port** field when adding the NF, or change the **Default NETCONF Port** in Settings.

If port 830 is blocked by a firewall, the connection test fails at the **TCP Connection** stage.

13.4 NETCONF over SSH

NETCONF itself does not provide encryption or login, so it runs on top of a secure transport. MiniSMO uses SSH (RFC 6242):

1. A normal SSH connection is established and the user logs in.
2. The client requests the SSH **subsystem** named `netconf`, instead of a shell.
3. On that channel both sides send `<hello>` messages, then exchange RPCs.

Messages are delimited ("framed") in one of two ways:

- **NETCONF 1.0**: each message ends with the marker `]]>]]>`.
- **NETCONF 1.1**: *chunked framing*, where each chunk is preceded by its length.

MiniSMO supports both and uses 1.1 when the device advertises it.

On first connect, MiniSMO records the device's SSH host key fingerprint. Later connections must present the same key, which protects you from connecting to an impostor. Changing the NF's host or port clears the pinned key.

14. Testing an NF Connection

14.1 NETCONF Connection Test

TEST CONNECTION runs the whole NETCONF connection pipeline and reports each stage separately, so you can see exactly where a problem is:

1. **Host Resolution:** turns the host name into IP addresses.
2. **TCP Connection:** opens a TCP connection to the NETCONF port.
3. **SSH:** SSH protocol handshake and server host key.
4. **Authentication:** logs in with the username and password.
5. **NETCONF subsystem:** asks SSH for the `netconf` subsystem.
6. **Server Hello:** exchanges `<hello>` messages, reads the session ID and the server capabilities, and agrees the NETCONF version (1.1 if both sides support it, otherwise 1.0).

MiniSMO shows **CONNECTED** only after a real NETCONF session has been established, meaning the server hello was received and parsed. A successful TCP or SSH connection alone is not enough.

The test uses the timeouts from *SYSTEM > Settings* (NETCONF Connect Timeout, default 10 seconds). After the test, MiniSMO closes the session with `<close-session/>`. Test sessions are recorded in the NETCONF Log with the NF name prefixed by `TEST`.

For a saved NF, the test also checks the SSH host key against the one pinned on first connect. If the key changed, the SSH stage fails with *host key changed*. If the device was legitimately replaced or re-keyed, use **FORGET HOST KEY** on the NF's Overview tab and connect again.

Related: [The Hello Exchange](#), [NETCONF Connection Troubleshooting](#).

14.2 The Hello Exchange

Immediately after the `netconf` subsystem opens, both sides send a `<hello>` message:

- The **server hello** lists every capability the device supports and contains a `<session-id>`.
- The **client hello** from MiniSMO advertises `urn:ietf:params:netconf:base:1.0` and `urn:ietf:params:netconf:base:1.1`.

If both sides list base 1.1, the session uses 1.1 chunked framing; otherwise it uses 1.0.

In MiniSMO the hello exchange is the last stage of **TEST CONNECTION** and of **CONNECT**. The Server Hello stage fails if:

- the device does not send a hello within the connect timeout,
- the hello is not well-formed XML,

- the hello lists no capabilities or no NETCONF base capability.

Both hellos are recorded in the NETCONF Log with operation `hello`.

15. Understanding NETCONF Capabilities

The **capabilities** in a device's hello describe what it can do. MiniSMO stores them each time you connect and shows them on the NF's **Capabilities** tab.

Column	Meaning	Example
Capability URI	The full URI as sent by the device	<code>urn:ietf:params:xml:ns:yang:ietf-interfaces?module=ietf-interfaces&revision=2018-02-20</code>
Module	YANG module name, if the URI names one	<code>ietf-interfaces</code>
Revision	YANG module revision date, if present	<code>2018-02-20</code>
Parameters	Remaining URI parameters, such as features or deviations	<code>features=ntp</code>

Protocol capabilities such as

`urn:ietf:params:netconf:capability:candidate:1.0` are shown with a module of `netconf:candidate` and a parameter `version=1.0`.

Use the search box (Ctrl+F) to filter by any part of the URI, for example `o-ran`, `ietf-hardware` or `candidate`.

Capabilities tell you:

- which YANG modules (and revisions) the device implements, so you know which models to import into the [YANG Model Registry](#),
- which datastores exist (`:candidate` , `:startup`),
- optional protocol features (`:with-defaults` , `:xpath` , notifications, and so on).

16. Running GET

`<get>` asks the device for its **running configuration plus operational state** (counters, status, hardware inventory and so on).

1. Open *FCAPS > Configuration* and select the NF in the list.
2. Open the **NETCONF read** tab.
3. Optionally enter a **subtree filter** (an XML fragment) to limit the reply, for example:

```
<interfaces-state xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces"/>
```

4. Select **GET**.

If the NF is not connected, MiniSMO connects first. The result panel shows:

Field	Meaning
Status	OK, or ERROR with the device's error message
Message ID	The <code>message-id</code> of the RPC
Session ID	The NETCONF session ID
Timestamp	When the request was sent
Duration	Round-trip time in milliseconds
Request / Response	Full XML, as an expandable tree or raw text

Unfiltered `<get>` on a large device can return several megabytes and take a while; use a filter if you know what you need. The NETCONF Operation Timeout in Settings (default 30 seconds) applies.

For simulated NFs, GET returns synthetic XML with session ID `SIM`.

17. Running GET-CONFIG

17.1 Running GET-CONFIG

`<get-config>` returns **configuration only** (no operational state) from a named datastore.

1. In *FCAPS > Configuration*, select the NF and open the **NETCONF read** tab.
2. Choose the **Datastore** in the toolbar above the tabs. Only datastores the device advertised are enabled: *running* always, *candidate* if the device has the `:candidate` capability, *startup* if it has `:startup`.
3. Optionally enter a subtree filter.
4. Select **GET-CONFIG**.

For external NFs, the **Parameters**, **XML tree** and **Raw XML** tabs of the same page show the running configuration, as an XML tree and as raw XML.

Related: [Running, Candidate and Startup Datastores](#).

17.2 Running, Candidate and Startup Datastores

Datastore	What it holds	Always present?
running	The configuration the device is using right now	Yes
candidate	A scratch copy that can be edited and then committed to running	Only with <code>:candidate</code>
startup	The configuration loaded at boot	Only with <code>:startup</code>

Why might a device not support a datastore? Candidate and startup are optional in NETCONF. Many O-RUs, for example, apply changes directly to running and keep it persistent, so they advertise neither. MiniSMO greys out datastores that a device has not advertised; if you try one through the API anyway you get *DATASTORE_NOT_SUPPORTED*.

Simulated NFs support running and candidate.

18. Viewing Raw NETCONF XML

Everywhere MiniSMO shows a NETCONF exchange, you can switch between:

- **Tree:** an expandable view of the XML elements with their values. Good for browsing large replies.
- **Raw XML:** the exact text sent or received, pretty-printed for reading.

Select **COPY** to put the raw XML on the clipboard, for example to paste into a ticket or compare with another tool.

The same raw XML is stored in the NETCONF Log. Passwords never appear in NETCONF XML; authentication happens in SSH, before any NETCONF message is exchanged.

19. NETCONF Log

LOG > NETCONF Log records every NETCONF message MiniSMO sends or receives, for external and simulated NFs.

Column	Meaning
Timestamp	Local time the message was logged
DIR	OUT (MiniSMO to device) or IN (device to MiniSMO)
NF	NF name (TEST <name> for connection tests)
Operation	For example hello, get, get-config, get-schema, create-subscription, establish-subscription, notification, edit-config, commit, retrieve-file-list, file-upload, close-session
Message ID	RPC message-id (empty for hello)
Session ID	NETCONF session ID (SIM for simulated NFs)
Status	SENT for requests; OK or ERROR for replies; BLOCKED for a request refused before it was sent (see Write Protection)
Duration	Round trip in ms, on replies

Select a row to see the full XML below the table. The list refreshes every few seconds while the page is open; new messages appear without moving your scroll position or the selected row.

- **Search** (Ctrl+F) matches NF name, operation, message ID and XML content.
- **Filters:** Operation (hello, get, get-config, close-session), Direction (OUT, IN) and Status (OK, ERROR, SENT).
- **COPY** copies the selected message's XML.
- **EXPORT** asks for confirmation, saves the current view as JSON or CSV to `%LOCALAPPDATA%\Kaldrivon\MiniSMO\exports` and shows **Open file / Show in folder**.
- **CLEAR** deletes all log entries after confirmation.

Messages are kept for 7 days by default, optionally with a size limit, and very large payloads are capped in history and marked *PAYLOAD TRUNCATED FOR HISTORY*. See [NETCONF Log Retention](#). Passwords are never logged.

20. Monitoring External NFs

20.1 Monitoring External NFs

The **Monitoring** tab of each external NF (*NETWORK > Network Functions*, select the NF) collects alarms (FM), configuration changes (CM) and performance data (PM) from external NETCONF devices. Monitoring observes, reads and subscribes. Only two things change an NF's configuration: pointing it at MiniSMO's VES collector (see [Pointing an NF at the Collector](#)) and configuration changes you enable and confirm for that NF (see [Changing External Configuration](#)). A third request acts on the NF without changing its configuration: a measurement file upload you confirm under **FILES ON NF...** (see [Measurement Files](#)).

When you add an NF under *Network Functions* (see [Adding an External NF](#)), MiniSMO integrates it straight away: it connects, discovers the capabilities, starts FM, CM and PM monitoring with the best method each function supports, and points the NF's VES destination at the collector. A dialog shows each step as it completes. Turn this off in *SYSTEM > Settings > General*, section **Integration**, if you prefer to start by hand.

To start by hand, select the NF in *Network Functions*, open its **Monitoring** tab and choose **DISCOVER** (MiniSMO reads the server `<hello>`, the YANG library and the event stream list; see [YANG Library Discovery](#)), then **START MONITORING**, pick FM, CM and/or PM, and confirm. The same button changes the settings of an NF that is already monitored.

MiniSMO opens a separate NETCONF session for notifications, reads the current state (active alarms, a configuration snapshot) and subscribes. If the device does not answer one of these initial reads (some simulators never reply to a read of the O-RAN active alarm list), that function is marked *initial sync incomplete* with the reason; the session, the subscription and the other functions carry on.

The tab also has a **VES** card: whether the NF's VES destination points at MiniSMO, how many VES events have arrived, the last event, the heartbeat and the last pnfRegistration. It shows what the collector actually receives, not what the NF's models say it could send. An NF only sends VES when its own VES reporting is on (for O-RAN-SC NTS simulators, the simulator's VES features) and it can reach this PC. The subscription lives only as long as that session. Nothing is stored on the device, and **STOP** just closes the session. While monitoring of an NF is stopped, its VES events are still listed in the VES event log (status NOT_MONITORED) but create no alarms and no PM. **STOP** ends FM, CM, PM and VES processing for the NF. The **MONITORING** column in *Network Functions* shows ONGOING, CONNECTING, STARTING, RECONNECTING, FAILED, STOPPED or NOT MONITORED for each external NF.

Supervision and alarm resync. When the monitoring session of an NF is down for more than 30 seconds, MiniSMO raises the internal alarm **Loss of supervision** on that NF and clears it when the session is back. On reconnection MiniSMO reads the

NF's active alarm list again and reconciles the alarm view: alarms no longer active on the NF are cleared. For an NF that reports its faults by VES and keeps a 3GPP AlarmList, a VES alarm whose alarmId is no longer in the list is cleared too (after a one-minute grace period, because a new fault may not be in the list yet).

RESYNC ALARMS (on the Monitoring tab and in the Network Functions toolbar) does the same on request. If the NF's alarm list cannot be read (some NFs only send notifications), MiniSMO tells you so and offers to clear the NF's active alarms locally; faults still active on the NF come back with its next notification or VES event. Nothing is sent to the NF.

O-RAN alarms (o-ran-fm). An `alarm-notif` is mapped field by field: the title is `fault-text`, the faulty object is `fault-source` (the `affected-objects` are listed in the details), the severity is `fault-severity`, `is-cleared` = true closes the matching alarm, and the time is `event-time`. `fault-id` is shown only as the alarm ID. One alarm instance is NF + `fault-id` + `fault-source`, because the same fault-id can be active on several sources. Many O-RUs only send these notifications and keep `active-alarm-list` empty or unreadable; the alarm view then follows the notification stream.

What you asked to monitor is stored locally. When MiniSMO restarts, it reconnects and subscribes again.

Screenshot placeholder: **Monitoring: connection health and FM / CM / PM state per external NF**

To be captured from the Windows 0.4.0 build (docs/images/screenshots/subscriptions.png) before public release.

20.2 Capability Discovery

Discovery shows, per function, what the device advertises and what MiniSMO will do with it. Everything is taken from the NF: its `<hello>`, its YANG library (RFC 8525 / 7895) and its event streams. Nothing is assumed per vendor or per NF type, so any O-RU, O-DU or O-CU that implements these standards can be monitored.

Function	Methods, in the order MiniSMO prefers them	Needs
FM	1. Alarm notifications of each alarm model the NF has	<code>notification:1.0</code> (RFC 5277) or <code>ietf-subscribed-notifications</code> (RFC 8639), plus <code>o-ran-fm</code> , <code>ietf-alarms</code> (RFC 8632) or the 3GPP AlarmList with RFC 6470 <code>netconf-config-change</code>
FM	2. Reading each model's active alarm list periodically	the alarm model alone

Function	Methods, in the order MiniSMO prefers them	Needs
CM	1. Change notifications	<code>ietf-netconf-notifications</code> (RFC 6470)
CM	2. YANG-Push on-change	<code>ietf-yang-push</code> with the on-change feature
CM	3. Reading the configuration periodically	nothing beyond <code><get-config></code>
PM	All of these side by side	YANG-Push periodic (RFC 8641); the NF's own PM model read with <code><get></code> (<code>o-ran-performance-management</code> for O-RUs, O-RAN-SC <code>nts-cell-pm</code>); O-RAN <code>measurement-result-stats</code> notifications; VES <code>perf3gpp</code> ; measurement files

The 3GPP AlarmList is also found by a read-only `get-config` probe, because some NFs (O-RAN-SC NTS) keep it in their configuration without announcing the module.

Namespaces come from the NF. Every read uses the namespace the NF announced for the module (for example a later version of `urn:o-ran:fm`). The standard namespace is used only for a module the NF did not announce. Notifications are recognized by module, so a newer version of the same module is still understood.

When an NF offers nothing for a function, the function is shown as **Unsupported** with the reason. Devices that report through VES are served by the [VES Collector](#). File-based PM is described under [Measurement Files](#).

20.3 Method Choice and Fallback

Automatic (default). For FM and CM, MiniSMO uses the first method in the table above that the NF offers. For PM, every source the NF offers runs at the same time, because they usually carry different data: interface statistics by YANG-Push, the NF's own PM model, VES streaming, measurement files. Values that arrive twice for the same measurement, object and period are stored once.

Fallback. When the method in use stops working, MiniSMO switches to the next one the NF offers:

- the event stream is refused: alarms are read from the NF's alarm list every 2 minutes, and the configuration is read every 15 minutes (or at the intervals you set);
- an alarm or configuration read fails 3 times in a row: the next method is tried;
- a YANG-Push subscription is refused: the same data is read with `<get>` at the same period (with the XPath when the NF supports XPath filters, or the equivalent subtree filter for the default selection);

- a PM model read fails 3 times: that source is marked failed and retried every few minutes; the other PM sources continue;
- O-RAN PM notifications stop arriving (usually because measurements are not activated on the O-RU): the O-RU's counters keep coming from the `<get>` read.

Every change is listed on the NF's **Monitoring** tab under **METHODS & FALLBACKS** (what failed, why, what is used now) and appears in the notifications (bell). A source that works again is listed too. After a reconnection MiniSMO starts again from the preferred method.

Choosing yourself. START MONITORING (also on an NF that is already monitored) lets you pick the FM and CM method from the ones this NF offers, turn single PM sources off, and turn automatic fallback off. A method the NF does not offer is not used, and that is listed. With fallback off, a method that stops working is shown as **Failed** with the reason instead of being replaced.

Discovery also reads the RFC 5277 stream list: stream names and whether the device keeps a **replay log**. Replay lets MiniSMO recover events missed during a disconnection.

21. Write Protection

MiniSMO changes external NFs in two ways only: the VES destination (set when the NF is added, unless turned off in *SYSTEM > Settings > General*, section **Integration**) and configuration changes you have enabled and confirmed for that NF (both described below). One more request acts on the NF: a measurement file upload (`file-upload`) that you confirm. It changes no configuration. The guarantee sits in the NETCONF layer: every request is checked against a short list of allowed operations before it leaves the application.

Allowed	Why
<code>get</code> , <code>get-config</code> , <code>get-schema</code> , <code>get-data</code>	Reading state and configuration
<code>create-subscription</code> (RFC 5277)	Transient notification subscription, ends with the session
<code>establish-subscription</code> , <code>delete-subscription</code> (RFC 8639/8641)	Dynamic subscriptions MiniSMO created itself
<code>retrieve-file-list</code> (o-ran-file-management)	Lists files on the NF, changes nothing
<code>file-upload</code> (o-ran-file-management)	Only after you confirm REQUEST UPLOAD... Asks the NF to push one file to your upload target. Recorded in the audit log
<code>close-session</code>	Ending its own session

Everything else is refused and not sent, unless it is part of one of the two changes below. That includes `edit-config`, `copy-config`, `delete-config`, `commit`, `discard-changes`, `lock` / `unlock`, `kill-session`, `reset`, `reboot`, software installation, and alarm acknowledgement or clearing. `copy-config`, `delete-config`, `kill-session`, `reset`, `reboot` and software installation are never sent at all. A refused request appears in the NETCONF Log with status BLOCKED.

The backend API refuses these operations too, so a script calling the API directly cannot write either.

Alarms of external NFs cannot be cleared or acknowledged from MiniSMO. You can add a **local note** or a **local acknowledgement**. Both are labelled *local only*, are stored in MiniSMO and are never sent to the NF.

MiniSMO also does not create configured (persistent) subscriptions, activate measurement jobs or set up file upload on the device. When a function needs one of these, MiniSMO says so and leaves it to you.

Exception 1: the VES destination. To receive VES events, an NF has to know the collector's address. When you add the NF, MiniSMO writes it into the NF's own VES

settings, unless **Point new NFs at the VES collector** is off in *SYSTEM > Settings > General*, section **Integration**:

- The integration dialog after **SAVE** shows the result of this step.
- The write goes through a separate check that allows only MiniSMO's own 3GPP `NtfSubscriptionControl` entry under the NF's `ManagedElement` (recipient address, notification types, heartbeat period), and only deleting that same entry. Any other content, another manager's entry, a replace operation or a second container is refused before it is sent.
- MiniSMO reads the settings back afterwards to verify them. Both the write and the read appear in the NETCONF Log, with passwords masked.
- MiniSMO writes again only when the NF's values differ: after a `pnfRegistration`, after a lost monitoring session comes back, or when you change the collector settings. See [Pointing an NF at the Collector](#).

Exception 2: configuration changes you enable. See [Changing External Configuration](#). Changes are off for every NF until you enable them for that NF, and each change is previewed and confirmed.

The generic write API stays closed.

22. Subscription States and Gaps

Connection health and data freshness are shown separately. A quiet alarm stream is not a fault.

State	Meaning
Connected	The notification session is up
Subscribed	The subscription is active; no events recently (normal for alarms)
Receiving data	Events arrived in the last 15 minutes (FM/CM) or within the expected interval (PM)
Stale	PM data expected but not arriving (more than three intervals). The reason says what to check
Reconnecting	The session dropped. MiniSMO retries with increasing waits (1 s up to 5 minutes, with random jitter) and subscribes again
Unsupported	The device offers no method MiniSMO can use for this function
Polling fallback	You enabled explicit polling because push is unavailable. Polling is never shown as a subscription
Failed	The device refused the subscription; the reason shows its error message

Gaps. When the session drops and the device has a replay log, MiniSMO asks for the missed events from the time of the last event it received. The gap is shown as *recovered*.

Without replay, the gap is recorded as *not recoverable*. Current alarms and configuration are resynchronized from the device, but events inside the gap are not available. Gaps are also recorded if events arrive faster than they can be processed.

Duplicates (for example events sent again during replay) are recognized and counted, not stored twice.

23. YANG Library Discovery

NETCONF 1.1 devices often list only a few modules in their `<hello>`. The full module set is in the YANG library, `ietf-yang-library` (RFC 8525), announced with the `yang-library:1.1` capability and its `content-id`. MiniSMO reads the library with a read-only `<get>`, falling back to the older `modules-state` (RFC 7895). Modules learned this way appear in the NF's capability list, marked as coming from the YANG library.

The module set drives what MiniSMO does per NF:

Module	Effect
<code>_3gpp-common-fm</code>	FM from the 3GPP AlarmList (pull, plus change notifications if the NF sends them)
<code>_3gpp-common-measurements</code>	PM jobs live on the NF; values arrive as VES perf3gpp
<code>_3gpp-common-subscription-control</code>	VES destination set with an <code>NtfSubscriptionControl</code> entry (with your confirmation)
<code>o-ran-ves-subscribed-notifications</code>	O-RU: VES destination set with a configured subscription (RFC 8639), heartbeat via <code>o-ran-supervision</code>
Any module with <code>config=false</code> counter containers	PM pull, found by PM discovery in the NF's own YANG (for example a cell PM module)
<code>ietf-subscribed-notifications</code> , <code>ietf-yang-push</code>	Dynamic subscriptions and YANG-Push

A missing module never breaks monitoring. The function is shown as unsupported, with the reason. When the NF reports a new library `content-id`, or sends a capability or library change notification, MiniSMO reconnects and discovers again.

24. The VES Collector

24.1 VES Collector

Many O-RAN and 3GPP NFs report through VES (the ONAP / O-RAN event listener format, version 7.2.1) instead of, or next to, NETCONF notifications. MiniSMO includes a VES collector: an HTTPS (and optionally HTTP) listener that accepts `POST /eventListener/v7` and `/eventListener/v7/eventBatch`.

It runs on its own ports. The MiniSMO API itself stays bound to this PC only. Its event list, NFs and unattributed sources are on *SYSTEM > Settings > VES Collector*. The listener settings are in *SYSTEM > Settings > General*, section **VES collector**; **COLLECTOR SETTINGS** on the VES Collector tab opens them, and **RESTART COLLECTOR** restarts the listener. The VES state of one NF is also on that NF's **Monitoring** tab.

Setting	Default
Listen address	<code>0.0.0.0</code> (all network interfaces)
HTTPS port	8443, with a self-signed certificate created on first start
HTTP port	8080 (can be switched off)
Authentication	none, or HTTP Basic with a username and password you set

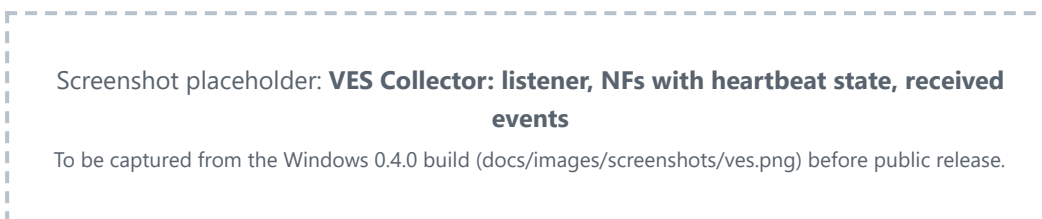
The installer adds a Windows Firewall rule for TCP 8443 and 8080. If you choose other ports, the VES Collector tab shows the `New-NetFirewallRule` command to run in an administrator PowerShell. The certificate's SHA-256 fingerprint is shown too, so you can compare it with what the NF reports.

What the collector does with each domain:

Domain	Result
<code>fault</code>	Raise, update or clear an alarm. Title <code>specificProblem</code> (else <code>alarmCondition</code> , else <code>eventName</code>), category <code>alarmCondition</code> , object <code>alarmInterfaceA</code> . The identity is the NF (<code>sourceName</code>), <code>alarmInterfaceA</code> and <code>alarmAdditionalInformation.alarmId</code> (the condition when there is no alarm ID). <code>eventName</code> is a category, never an identity. <code>NORMAL</code> clears the matching alarm

Domain	Result
perf3gpp	PM samples. Every <code>measInfo</code> has its own ordered <code>measTypes.sMeasTypesList</code> ; a <code>measResults</code> entry's <code>p</code> is a 1-based index into it, so each value is stored under its measurement name (for example <code>RRU.PrbUsedDL</code>), per <code>measObjInstId</code> , stamped with the period end (<code>lastEpochMicrosec</code>) and the <code>granularityPeriod</code> . A value stamped more than 5 minutes (or one period, if longer) after this PC's clock cannot have been measured yet: it is refused, counted in the VES log entry, and the NF's PM source explains how far ahead the NF's timestamps are. Check the NF's clock and the measurement period it reports. <code>suspectFlag "true"</code> marks values as suspect in charts; a <code>p</code> without a name is dropped
heartbeat	Heartbeat state per NF: OK while heartbeats arrive within three intervals, otherwise MISSED
pnfRegistration	Registration details (vendor, model, serial, OAM address). Counts as "the NF is up": MiniSMO resubscribes and, if you opted in, re-points the NF at the collector
stndDefined	3GPP FaultSupervision (<code>notifyNewAlarm</code> , <code>notifyChangedAlarm</code> , <code>notifyClearedAlarm</code>) becomes alarms. Other namespaces are stored and shown as not interpreted
other domains	Stored and shown as not interpreted

Invalid requests get `400` with a VES `requestError`. Wrong or missing Basic credentials get `401`. Accepted events get `202`. The same event received twice is stored once as data and marked *duplicate* in the event list.



24.2 Matching Events to NFs

A VES event names its sender in `commonEventHeader`: `sourceName`, `sourceId` and `reportingEntityName`. MiniSMO matches these, case-insensitively, against:

1. the **VES source names** you entered for an NF (VES Collector tab, **Other VES source names for this NF**, then **SAVE**),
2. the NF's name in the inventory,
3. as a last resort, the sender's IP address when exactly one external NF has that host address, and only if that `sourceName` was never attributed by name to a different NF (containers on one host share an address: the events of a deleted O-DU must not land on the O-RU next to it). Otherwise the event is listed as unattributed, where you can assign it.

An event that matches no NF is kept as **unattributed**. The page lists these sources with the address they came from. **ASSIGN...** adds the source name to an NF, so its later events are attributed. Events received before that stay unattributed.

Tip: an O-RAN-SC NTS device sends its own hostname or NF name as `sourceName` . Name the NF the same in MiniSMO, or add that value as a VES source name.

25. Pointing an NF at the VES Collector

An NF only sends VES events once it knows where to send them. MiniSMO writes the collector address into the NF's own VES settings automatically when you add the NF, so there is nothing to do on the *VES Collector* tab.

MiniSMO uses only the standard way to tell an NF where to send notifications: a **3GPP** `NtfSubscriptionControl` entry under the NF's `ManagedElement` (3GPP TS 28.623, module `_3gpp-common-subscription-control`). It checks that the NF announces this module in its `YANG library`, reads the NF's `ManagedElement` id and its existing subscription entries, and writes one entry of its own (id `MiniSMO`) with:

Leaf	Value
<code>notificationRecipientAddress</code>	the collector URL, e.g. <code>https://192.168.1.20:8443/eventListener/v7</code>
<code>HeartbeatControl/heartbeatNtfPeriod</code>	the heartbeat period (default 60 s; setting <code>ves_heartbeat_period_s</code> , 0 = no heartbeats)

O-RUs use the O-RAN WG4 M-Plane way (clause 18): a *configured subscription* (RFC 8639, `ietf-subscribed-notifications`) whose receiver uses the VES transport of `o-ran-ves-subscribed-notifications`. MiniSMO writes one subscription with the O-RU's `NETCONF` stream (or the first stream it lists), transport `o-ran-ves-connectionless` (no `encoding` leaf: RFC 8639 allows it only for configurable-encoding transports, and VES is JSON by definition) and one receiver named `MiniSMO` with `notification-recipient` = the collector's server root (e.g. `https://192.168.1.20:8443`; the O-RU may add path elements, and the collector accepts events at the root and at any path ending in `/eventListener/v7`). If the O-RU supports non-persistent M-Plane (`o-ran-supervision event-collector-monitoring`), MiniSMO also adds itself as heartbeat recipient with the heartbeat interval (at most 255 s, the model's limit). An NF that offers both models uses the 3GPP one.

Nothing device-specific is written, and only that one entry may be sent (MiniSMO checks the XML before it leaves). Entries of other managers are never changed or removed; a device that holds a single entry and has it in use refuses MiniSMO's, and the message says so. The standard model has no credential fields: if the collector requires basic authentication, set its user name and password on the NF itself (the preview warns about this). An NF without the model is reported in the integration step and nothing is written; set its destination with the vendor's tools, and the collector accepts its events anyway.

MiniSMO sends the `edit-config` to the running datastore (or the candidate plus `commit` if the NF has no writable running), then reads the entry back and reports whether it matches. The result is shown in the integration dialog after **SAVE**.

The address used is the local address this PC uses to reach that NF, plus the collector port for the protocol chosen in Settings. If NFs reach this PC through NAT or a DNS

name, set **Collector address NFs should use** in *SYSTEM > Settings > General*, section **VES collector**.

MiniSMO keeps the NF pointed here: it re-applies the same settings after the NF re-registers (`pnfRegistration`), after a lost monitoring session comes back, and when you change the collector address, port, protocol or credentials. It writes only when the NF's entry differs, and only MiniSMO's own entry.

To stop MiniSMO from writing the VES destination, turn off **Point new NFs at the VES collector (and keep them pointed here)** in *SYSTEM > Settings > General*, section **Integration**.

Every write and every read-back is in the NETCONF Log. Passwords are masked there and in the application log.

When you delete an NF, the delete dialog has the option **Stop this NF sending VES to MiniSMO** (on by default). With it, before the NF is removed MiniSMO deletes its own `NtfSubscriptionControl` entry (or, on an O-RU, its configured subscription and heartbeat recipient), so the NF stops sending here. Other managers' entries stay as they are. The result is shown after the deletion. If the NF cannot be reached, the NF is still deleted and the message says so.

Events that still arrive from a deleted NF (one that could not be changed, or one whose destination you set by hand) are **dropped, not stored**. The *VES Collector* tab lists them under **DELETED NFs STILL SENDING**, and the counters show how many were dropped. Adding an NF with the same name or address again accepts its events again; **ACCEPT THEIR EVENTS AGAIN...** does that for all of them (they are then stored as unattributed).

26. Fault Management

26.1 Fault Management

FCAPS > Alarms opens the **Alarm Console**. It lists simulated, external and internal alarms together. The counts above the list show active alarms by severity.

Column	Meaning
SEVERITY	CRITICAL, MAJOR, MINOR, WARNING, or CLEARED once cleared
NF	The NF the alarm is raised on
ALARM / DESCRIPTION	Alarm title and text
ORIGIN	Simulated, External or Internal
RAISED	When the alarm was raised

Filters: alarm state **Active** (default), **Cleared** or **All**; severity; origin (Any origin, Simulated, External, Internal); NF. The search box (Ctrl+F) matches alarm, text, NF and source.

Select an alarm to see **ALARM DETAILS**: source, managed object, times, the **EVENT HISTORY** and, for external alarms, a bounded copy of the last notification. You can add a **LOCAL NOTE (SAVE LOCAL NOTE)** and tick **Acknowledged locally**; both stay in MiniSMO and are never sent to the NF. **INVESTIGATE** (Pro) is described under [Investigate](#).

EXPORT CSV asks for confirmation, writes all alarms (active and history) to a CSV file in the exports folder and shows **Open file** / **Show in folder**.

RAISE TEST ALARM... (at the top of the Alarm Console): choose a simulated NF, an alarm type and a severity, optionally add text, and select **RAISE ALARM**. If the same alarm type is already active on that NF, it is updated instead of duplicated.

CLEAR (SIMULATED ALARM): select an active simulated alarm and choose **CLEAR (SIMULATED ALARM)**. It stays in the list with severity CLEARED and a cleared timestamp.

The worst active severity per NF is also shown on the Topology view, and the Dashboard counts active, critical and major alarms.

Alarms can only be generated on simulated NFs. Alarms of external NFs arrive by themselves, through NETCONF notifications, alarm-list reads or VES fault events. See [External Alarms](#).

Internal alarms (origin *Internal*, source MINISMO) are raised and cleared by MiniSMO itself:

Alarm	Raised when	Cleared when
Loss of supervision (MAJOR, on the NF)	the monitoring session of the NF is down for more than 30 s	the session is connected again, monitoring is stopped, or the NF is deleted
VES heartbeat missed (MINOR, on the NF)	an NF that sends VES heartbeats misses three intervals	heartbeats arrive again, monitoring is stopped, or the NF is deleted
Storage usage high, High memory usage, High CPU usage, Notification queue backlog, Events dropped by the event bus, Operations jobs failed, Webhook deliveries failed (on <i>MiniSMO</i>)	the matching System health check is degraded or critical in two consecutive one-minute readings	the check is healthy in two consecutive readings
VES collector not listening (MAJOR)	a VES listener could not start	it is listening

Use the *Internal* origin filter to see only these. The supervision alarms (Loss of supervision, VES heartbeat missed) belong to the NF: they count in alarm correlation and in *Most alarms* on *TOOLS > Analytics*. The health alarms are raised on MiniSMO itself and are left out of both.

For external NFs, **RESYNC ALARMS** reads the NF's alarm list again. See [Monitoring External NFs](#).

26.2 About Simulated Alarms

The simulator knows ten alarm types, loosely based on what real RAN equipment reports:

Alarm type	Typical meaning on real equipment
VSWR High	Antenna or feeder mismatch, reflected power too high
Cell Unavailable	Cell out of service
Loss of Signal	No RF output or no input on a link
Synchronization Lost	Loss of PTP/SyncE or GNSS timing
High Temperature	Unit above its temperature threshold
Interface Down	Fronthaul or transport interface down
Power Supply Failure	PSU or input power fault
CPU Load High	Processing load above threshold
Memory Usage High	Memory above threshold

Alarm type	Typical meaning on real equipment
High PRB Utilization	Radio resources near saturation (congestion)

Severities follow the usual X.733-style scale: CRITICAL, MAJOR, MINOR and WARNING. A cleared alarm has severity CLEARED.

Scenarios raise and clear alarms automatically; their Source is `SCENARIO:`
`<scenario>`. Stopping or resetting a scenario clears the alarms it raised, but not alarms you generated manually.

All alarm data is synthetic.

27. External Alarms

External alarms appear in the *Alarm Console* together with simulated ones. The **Origin** column and filter tell them apart.

Alarms reach MiniSMO in three ways, and all three go into the same alarm store:

- **Pull.** MiniSMO reads the NF's alarm list (`o-ran-fm`, `ietf-alarms` or a 3GPP `AlarmList`) and reconciles it.
- **NETCONF notifications.** RFC 5277 alarm notifications arrive in real time.
- **VES fault events.** Events received by the `VES collector` are keyed by NF, object DN (`alarmInterfaceA`) and alarm condition. `NORMAL` severity clears the alarm.

When monitoring starts, MiniSMO reads the device's active alarm list (`o-ran-fm active-alarm-list` or `ietf-alarms alarm-list`). This is the *initial synchronization*. If the list cannot be read, FM shows *sync incomplete*: alarms raised before monitoring started may be missing until they change.

After that, alarm notifications update the current alarms:

- A new alarm identity raises an alarm.
- The same identity with new severity or text updates it.
- A clear clears it.

The identity is the device's own key: `fault-id` + `fault-source` + affected objects for O-RAN, resource + alarm type + qualifier for `ietf-alarms`, or the object DN + `alarmId` for 3GPP alarms. `alarmId` alone is not unique, so it is always combined with the object. A 3GPP alarm seen both as a VES fault and in the NETCONF `AlarmList` (with its `objectInstance`) is therefore one alarm.

Names and labels come from the alarm itself, never from a built-in table:

Shown as	3GPP <code>AlarmList</code> / <code>stndDefined</code>	VES <code>faultFields</code>
Alarm (title)	<code>specificProblem</code> , else <code>alarmType</code>	<code>specificProblem</code> , else <code>alarmCondition</code> , else <code>eventName</code>
Category	<code>alarmType</code>	<code>alarmCondition</code>
Faulty object	<code>objectInstance</code> / <code>href</code>	<code>alarmInterfaceA</code>
Probable cause, raised time	<code>probableCause</code> , <code>alarmRaisedTime</code>	<code>alarmAdditionalInformation.probableCause</code> / <code>.alarmRaisedTime</code>
Device alarm ID	<code>alarmId</code>	<code>alarmAdditionalInformation.alarmId</code>

The alarm ID is shown as the device's ID, never as the title. When a device sends no descriptive text at all, the title says so ("Unspecified alarm").

Some devices (the O-RAN-SC NTS simulators, for example) keep a 3GPP AlarmList whose records carry only `alarmId` and severity, and describe the same alarms in VES faults. For an NF that reports by VES (it was pointed at MiniSMO's VES collector, or VES events have arrived from it), such a record updates or clears the matching VES alarm and is never shown as an alarm of its own. When an NF is added, MiniSMO points it at the VES collector *before* it reads the AlarmList for the first time, so these records never appear; any that were shown earlier are removed at the next AlarmList read. For an NF that has only the AlarmList, those records are its only FM data and are shown as "Unspecified alarm".

3GPP AlarmList devices. Some 3GPP NRM devices (O-RAN-SC NTS simulators, for example) keep alarms as `alarmRecords` inside the running configuration. They signal a change with a `netconf-config-change` that points at a record's `perceivedSeverity`. MiniSMO treats these as alarm events, not configuration changes: shortly after such a notification it re-reads only the AlarmList (subtree filter) and records the differences as raise, update or clear. A record whose severity is `CLEARED` is a cleared alarm. If a record changes several times between two reads, only the latest state is seen, but every change notification is still counted. Some of these devices publish only `alarmId` and severity; the other columns stay empty then. Such a record is merged with the NF's VES alarm of the same `alarmId` instead of appearing twice, and an AlarmList read never clears alarms that arrived by VES. When the AlarmList's `administrativeState` is LOCKED, the Monitoring tab says the list is in maintenance: its content may not reflect current faults.

Every event is kept in the alarm's **event history**: raise, update, clear, snapshot and resync. The history also keeps events that arrive late. An update older than the alarm's latest state is stored as *out of order* and does not overwrite newer state. A clear for an alarm MiniSMO never saw is stored as *clear unmatched*.

The details panel shows the device's event time next to the time MiniSMO received the event, and a copy of the notification (limited in size).

28. Configuration Management

28.1 Configuration Management

FCAPS > Configuration shows and edits NF configuration. What you can do depends on the NF's mode.

Simulated NFs (editable)

Select a simulated NF in the NF list. The fields available depend on its type:

Field	Range / format	O-RU / O-DU	O-CU-CP / O-CU-UP
Administrative State	UNLOCKED or LOCKED	✓	✓
Cell Local ID	0 – 16383	✓	
PCI	0 – 1007	✓	
NR ARFCN	0 – 3279165	✓	
Bandwidth	5 – 100 MHz (standard NR channel widths)	✓	
Tx Power	0 – 49 dBm	✓	
TAC	0 – 16777215	✓	✓
MCC	3 digits	✓	✓
MNC	2 or 3 digits	✓	✓
Software Version	Free text	✓	✓

The buttons work like a small candidate/commit workflow:

Button	What it does
GET	Reloads the running configuration from MiniSMO's database
VALIDATE & STAGE	Validates your changes and stores them as a pending candidate. Running is unchanged
APPLY	Makes the candidate the new running configuration and increases the revision number
REVERT	Discards a pending candidate; if there is none, rolls running back to the previously applied revision

Invalid values (for example PCI 1500) are rejected with a message naming the field and its allowed range.

Setting **Administrative State** to LOCKED and applying it makes the NF's operational state DISABLED; its traffic counters drop to zero. Unlock it to bring it back.

Configuration is stored in the local SQLite database and survives restarts. The Demo Network uses the test PLMN 001/01.

HISTORY shows every applied change for the selected NF: timestamp, revision, field, old and new value, and source. See [Configuration History](#).

External NFs

For an external NF the page reads the configuration with `<get-config>` and shows it in the same parameter form. Tabs: **Parameters**, **XML tree**, **Raw XML**, **History** and **NETCONF read**. The toolbar has:

Button	What it does
Datastore box and READ (GET-CONFIG)	Reads the configuration from the selected datastore
TAKE SNAPSHOT	Asks for confirmation, reads the running configuration and stores it as a new version if it changed
ENABLE CHANGES... / DISABLE CHANGES	Allows or stops configuration changes on this NF (off by default)
REVIEW CHANGES...	Shows the changed values and the exact <code>edit-config</code> , then APPLY TO NF
DISCARD EDITS	Drops your unsent edits

On the **History** tab, **EXPORT** asks for confirmation, writes the selected snapshot (secrets masked) to the exports folder and shows **Open file / Show in folder**. See [Changing External Configuration](#) and [How MiniSMO Writes to External NFs](#).

28.2 How MiniSMO Writes to External NFs

Monitoring only reads: connect, discover capabilities, `<get>`, `<get-config>`, `<get-schema>` and subscriptions. Free-form writes are never sent, because configuration changes on real equipment without schema validation, locking and rollback can take a live cell or fronthaul link down.

Two configuration writes exist, each limited to exactly what it needs:

- **VES destination.** When you add an NF, MiniSMO writes the collector address into the NF's own VES settings (only the leaves of that one container) and reads them back. Turn this off in *SYSTEM > Settings > General*, section **Integration**. See [Pointing an NF at the Collector](#).
- **Configuration changes you enable for an NF.** Each change is validated against the NF's YANG, reviewed and confirmed, then applied in a transaction with read-back. See [Changing External Configuration](#).

A third request acts on the NF without changing its configuration: **REQUEST UPLOAD...** under **FILES ON NF...** sends an o-ran-file-management `file-upload` after you confirm it, so the NF pushes one measurement file to your upload target. See [Measurement Files](#).

Simulated NFs are fully editable on the *Configuration* page. See [Configuration Management](#).

29. External Configuration History

For external NFs, *Configuration* shows the configuration read with `get-config` in the same parameter form as simulated NFs, which is editable once you enable changes for that NF. The **XML tree** and **Raw XML** views are also there.

The **History** tab lists stored **snapshots** (versions). A new version is stored only when the configuration actually differs. Snapshots are taken:

- when monitoring starts,
- after a change notification,
- when you choose **TAKE SNAPSHOT** (after confirmation),
- by explicit polling, if you enabled it.

Change notifications (RFC 6470) tell MiniSMO *which paths* changed and who changed them. They do not carry old or new values. MiniSMO reads a new snapshot a couple of seconds after a change notification, and at most once every 30 seconds while changes keep coming, so one snapshot can cover several changes. On 3GPP AlarmList devices, changes inside the AlarmList are alarm events (see *External Alarms*): they create no configuration history entries, and the AlarmList is left out of snapshots. **COMPARE** shows the net difference between two versions (changed, added and removed leaves, plus a text diff). It does not claim an exact event-by-event history.

Secret values such as passwords, keys and SNMP communities are replaced with `*****` in snapshots, diffs, exports and the NETCONF Log. A snapshot cannot be pushed back to the NF from **History**; changes go through [Changing External Configuration](#) or, with Pro, [Rollback](#). **EXPORT** asks for confirmation and saves a local copy of a snapshot.

30. Changing External Configuration

Configuration changes on an external NF are off until you allow them for that NF. In *Configuration*, select the NF and choose **ENABLE CHANGES...**, then confirm.

DISABLE CHANGES makes it read-only again.

What can be edited. The parameters are the NF's own running configuration (`get-config`), not a list built into MiniSMO. Names, list keys, types and constraints come from the NF's YANG modules. MiniSMO reads them with `get-schema` , or uses models you imported under *YANG Models*, and caches them locally. Next to each value you see its range, length, units or choices. These are never edited:

- list keys (they identify the entry),
- `config false` state,
- secrets (shown as `*****`),
- leaf-lists,
- the 3GPP AlarmList (alarm data).

When a module cannot be read, its parameters show no constraints and the NF alone validates them.

Making a change.

1. Edit values; the running value stays visible next to each field.
2. Choose **REVIEW CHANGES...** MiniSMO re-reads the running configuration and checks each value against the YANG constraints. It refuses a value that changed on the NF since you read it (someone else edited it). You then see each old → new value and the exact `edit-config` .
3. **APPLY TO NF** runs a NETCONF transaction: - With `:candidate` : lock the candidate, `edit-config` (merge), `validate` (if `:validate` is advertised), a **confirmed commit** (`:confirmed-commit`), read the running values back, then the confirming `commit` , and unlock. If a value does not read back as requested, MiniSMO cancels the commit (`cancel-commit`) or lets the NF roll back when the confirm timeout expires. If a step fails, the candidate is discarded. - With `:writable-running` only: lock running, `edit-config` with `test-then-set` (if `:validate:1.1`), read back, unlock.
4. The result lists every step. For a refused change it shows the NF's `rpc-error` as returned (`error-tag` , `error-path` , `error-message`).

The `edit-config` contains exactly the leaves you confirmed, plus the list keys that address them, merged into the existing configuration. MiniSMO checks the XML actually sent against the confirmed plan. Any other leaf, value or list entry, or any delete or replace operation, is refused before it leaves the application.

After a change, MiniSMO stores a new configuration version and a change entry by *MiniSMO (this PC)* in **History**. If you monitor the NF, its own `netconf-config-change` notification appears there too.

31. Performance Management

31.1 Performance Management

FCAPS > Performance shows performance data for simulated and external NFs. This section covers simulated NFs; for external NFs see [External Performance Data](#).

Select a simulated NF on the left. The page shows the latest value of each counter as a tile, and a chart of the selected counter. Choose the chart range: 15 minutes, 1 hour, 24 hours, 7 days, 30 days or 90 days. Short ranges use raw samples; longer ranges use 5-minute or hourly aggregates and show the min/max band around the average.

Counter	Unit	Notes
PRB DL %	%	Downlink physical resource block utilization
PRB UL %	%	Uplink PRB utilization
Active UEs	count	Connected users
DL Mbps	Mbit/s	Downlink throughput
UL Mbps	Mbit/s	Uplink throughput
CPU %	%	Processing load
Memory %	%	Memory utilization
Packet Loss %	%	Transport packet loss
Temperature	°C	Unit temperature

O-CU-CP reports no PRB or throughput values (it is control plane only).

Collection interval: choose 1, 5, 15 or 30 seconds. The live tiles update every second either way; the interval controls how often a sample is stored in history (and therefore the chart resolution). The interval is also available in Settings as *PM Update Interval*.

Retention: raw samples are kept for 6 hours, 5-minute aggregates for 7 days and hourly aggregates for 30 days by default (the MINIMAL preset). See [PM Retention](#).

RESET PM deletes all stored PM history for all NFs after confirmation. Live generation continues.

EXPORT CSV asks for confirmation, writes the samples to the exports folder and shows **Open file** / **Show in folder**.

31.2 About PM Simulation

The simulated counters are designed to look like a real network, not random noise:

- Each NF follows a slow load curve (a roughly 10-minute cycle with a shorter ripple), with a different phase per NF so cells don't move in lockstep.
- Small random variation is smoothed, so values drift instead of jumping.
- CPU, memory and temperature follow load more gently than traffic does.
- When a scenario changes a counter, the value moves towards the new level over a few seconds instead of snapping.
- A LOCKED or DISABLED NF carries no traffic: PRB, UEs and throughput go to zero.

The absolute values are plausible for a mid-band 100 MHz carrier but are synthetic and not tied to any vendor's equipment.

32. External Performance Data

32.1 External Performance Data

Performance lists external NFs next to simulated ones. For an external NF, pick the **Metric** (the NF's own measurement names, for example `RRU.PrbUsedDL`) and then the **Managed object**. Pick the object from the list or from the **Managed objects** tree, where objects are grouped under their parent in the DN (for example the `NRCeLLDU` cells under their `GNBDUFunction`, the O-DU). Groups expand and collapse with a click. Then choose a time range and, for counters, rate or raw value. Values the NF flagged as suspect (`suspectFlag`) are marked.

Source	Metrics
YANG-Push periodic	Every numeric leaf under the subscribed XPath (default: interface statistics). List keys identify the resource, for example <code>interface eth0</code>
O-RAN PM notifications	<code>measurement-result-stats</code> : transceiver statistics (min, max, first, latest per unit, with units such as dBm) and window counts (<code>RX_ON_TIME</code> , <code>TX_TOTAL</code> ...)
Pull from a PM model the NF describes	MiniSMO asks the NF which PM models it has: it reads the YANG of the NF's non-IETF modules (YANG library and <code><get-schema></code>) and offers every <code>config false</code> container of numeric leaves that is measurement data: its leaves use counter types (<code>Counter32</code> / <code>Counter64</code> ...), or the container or module is named as measurement data (<code>pm</code> , <code>counters</code> , <code>statistics</code> , <code>measurements</code> , <code>kpi</code>). Operational state with units (a sync time error, a temperature), capability values (delay profiles) and configuration are not PM and are left out. Only the leaves the YANG declares in that container are read. Units come from the YANG <code>units</code> statement; Counter types and counter-like names are treated as cumulative. These sources are marked <i>found in the NF's YANG</i> in the PM sources list and can be switched off one by one. Nothing about them is built into MiniSMO
Pull from the NF's PM model (presets)	For models MiniSMO knows how to read (for example O-RAN-SC <code>nts-cell-pm</code>): each <code>pm-counters</code> container, keyed by the managed object's DN, read with a <code><get></code> at the interval you set (default 5 minutes). Labelled PULL. A PM module usually only augments the 3GPP NRM (<code>nts-cell-pm</code> puts <code>pm-counters</code> under each <code>NRCeLLDU</code>), so MiniSMO never reads it from the module's own root. It resolves the data path through the NF's YANG augment targets, or the instance tree when the modules cannot be read, and filters from the managed-object root (<code>ManagedElement</code> / <code>GNBDUFunction</code> / <code>NRCeLLDU</code> / <code>pm-counters</code>) with the right namespace at each level. The Monitoring tab shows the path it uses

Source	Metrics
O-RU measurement results (O-RAN WG4)	For O-RUs with <code>o-ran-performance-management : performance-measurement-objects</code> read with <code><get></code> (the results are <code>config false</code> , so never with <code><get-config></code>), every 60 s by default. The namespace is the one the O-RU's YANG library gives. Each <code>rx-window-measurement-objects</code> entry is stored per object unit: RU (the ietf-hardware component, for example <code>chassis</code>), TRANSPORT (<code>transport <flow name></code>) or EAXC_ID (<code>eAxC <id></code>). Measurement names are shown readably with the enum kept, for example <code>Rx on time (RX_ON_TIME)</code> . These counts are cumulative, so the change between polls is shown (rate or raw). <code>transceiver-measurement-objects</code> give min, max, first and latest per <code>transceiver <unit id></code> ; an O-RU without transceiver entries simply has none. Only entries the O-RU returns are used. There is no 3GPP DN: the values belong to the NF and its object unit. <code>measurement-result-stats</code> notifications are stored as well when the O-RU sends them
Measurement files	3GPP <code>meas CollecFile</code> per granularity period, signalled by VES <code>fileReady</code> , 3GPP <code>notifyFileReady</code> or NETCONF file-upload notification, see Measurement Files . Labelled FILE
VES perf3gpp	3GPP TS 28.550 streaming PM received by the VES collector : each <code>measResults</code> value is matched to its <code>sMeasTypesList</code> name and stored per <code>measObjInstId</code> and granularity period
Polling fallback	Numeric leaves returned by the subtree filter you give, labelled POLL

For an external NF the page also offers **Range** (15m to 30d), **Show** (auto, value or rate) and **Compare with** another NF.

Counters (octets, packets, errors ...) are shown as a rate per second by default. A counter that goes down is marked as a **reset**, and no rate is drawn across it. Missing intervals are drawn as **gaps** (breaks in the line), never as zero. Late samples are stored but flagged *out of order*, and duplicates are ignored.

The freshness indicator shows when the last sample arrived and whether it is Receiving or Stale. **EXPORT CSV** asks for confirmation, writes the samples with their source, method and quality flags, and shows **Open file** / **Show in folder**.

External PM samples, including values from measurement files, are kept as long as the 5-minute PM aggregates (7 days by default) and are not aggregated. See [PM Retention](#).

O-RAN radio units only count measurements that are activated in their configuration (`performance-measurement-objects : active`, measurement intervals, notification interval). The Monitoring tab shows the settings MiniSMO read; it does not change them. If no results are returned or PM stays *Stale*, activate the measurements with the device's own tools. File-based PM is a separate mechanism, see [Measurement Files](#).

32.2 Measurement Files (File-based PM)

Bulk PM: the NF writes one 3GPP measurement file (`measCollectFile` , TS 32.432 / 32.435) per granularity period and signals that it is ready. This is separate from the live counters read with `<get>` . MiniSMO takes file names, locations, counters and object DNs from the signals and the file content; nothing about them is built in.

How a file reaches MiniSMO

1. **Collection is configured on the NF** (`o-ran-performance-management` : active measurements, intervals, and the upload target in `remote-file-uploads` , or the deprecated `remote-SFTP-uploads` , which is ignored when `remote-file-uploads` is present). MiniSMO reads and shows this configuration under **SETTINGS...**; it never changes it.
2. **The NF signals the file**, by any of: - a VES `notification` event with `changeType FileReady`: each `arrayOfNamedHashMap` entry gives the file name and its location . The NF is identified from `sourceName` / `sourceId` , like any VES event. - a VES `stdDefined 3GPP notifyFileReady` (`fileInfoList[]` . `fileLocation`). - a NETCONF `file-upload-notification` (`o-ran-file-management`) on the monitoring session's event stream.
3. **MiniSMO retrieves the file** from the location the NF gave: `sftp://` , `ftpes://` (FTP over TLS, encrypted data channel), `https://` or `http://` (credentials only over https). Plain FTP and redirects are not used.
4. **The file is parsed and stored**: for each `measInfo` , the `measType` positions are mapped to names (both the `<measType p="1">` and the space-separated `<measTypes>` forms), and each `measValue` gives the measured object (`measObjLdn`). Values are stored per NF, object, measurement and period end, with the granularity period. `suspect` values are marked. gzip files are unpacked.

Turning it on. On the NF's **Monitoring** tab, **FILE-BASED PM > SETTINGS...**: turn retrieval on and choose the credentials: the NF's NETCONF credentials (common on O-RU simulators) or a separate user name and password for its file server. Passwords are stored protected and never logged. Files are retrieved only from the NF's own address and the servers you list. Until retrieval is on, file-ready signals are recorded with the reason, and nothing is fetched.

SMO pull. FILES ON NF... sends `retrieve-file-list` for a logical path (O-RAN WG4 defines `o-ran/pm` for measurement files; use what your NF documents). It lists files and changes nothing. **REQUEST UPLOAD...** asks the NF to push the selected file to the **upload target** you set (your own `sftp://` or `ftpes://` server) with `file-upload` ; it is sent only after you confirm, changes no configuration, and is recorded in the audit log. The NF's `file-upload-notification` then names the file, which is retrieved like any other.

IMPORT FILE... stores a measurement file you have on disk for the NF (for example copied from its file server).

Correctness

- A file is ingested once. A file signalled again (same name for the same NF) is counted as a re-notification, not stored twice.
- Values are per period. They are never treated as cumulative across files.
- When the same measurement of the same object and period already arrived another way (VES `perf3gpp`), it is not stored twice; the file reports how many values were already stored.
- The NF is the one that signalled the file. The file's own `managedElement` (`localDn` , `userLabel` , `swVersion`) is shown with it.
- When a file cannot be retrieved (retrieval off, host not allowed, scheme not supported, server refused the credentials, file larger than 50 MB, not a measurement file), the reason is shown with the file and nothing else is affected. **RETRY FAILED** tries again.

Live counters (`<get>`) are for "right now"; measurement files are the history. Both are kept, with their method (PULL or FILE) on every value.

33. Running Scenarios

SIMULATION > *Scenarios* plays scripted situations on the Demo Network so you can watch how they show up in FM, PM and Topology. Scenarios only ever affect **simulated** NFs; external NFs are never touched. The Demo Network must exist before you can run one.

Scenario	What happens	Duration
NORMAL TRAFFIC	Baseline load with natural variation	Until stopped
MORNING TRAFFIC RAMP	Network-wide load climbs from 30% to 140% of normal	3 minutes, then completes
CELL CONGESTION	O-RU-003: DL PRB above 95%, more UEs, higher packet loss; <i>High PRB Utilization</i> (MAJOR)	Until stopped
CELL OUTAGE	O-RU-007 goes operationally DISABLED with no traffic; <i>Cell Unavailable</i> (CRITICAL) and <i>Loss of Signal</i> (MAJOR)	Until stopped
HIGH CPU LOAD	O-DU-001 CPU above 90%, memory and temperature up; <i>CPU Load High</i> (MAJOR)	Until stopped
INTERMITTENT LINK FAILURE	O-RU-004 fronthaul flaps every 8 seconds: DISCONNECTED/DISABLED with <i>Interface Down</i> , then back	Until stopped

Controls:

Control	Effect
RUN	Starts the selected scenario. If another one is running, it is stopped first
PAUSE	Freezes the scenario timeline. Its effects stay in place
RESUME	Continues a paused scenario
STOP	Ends the scenario, restores normal state and clears the alarms it raised
RESET	Stops any scenario, sets all simulated NFs back to ENABLED/CONNECTED and clears all scenario alarms

One scenario runs at a time. The status line shows the state (IDLE, RUNNING, PAUSED, STOPPED or COMPLETED) and elapsed time. Every run is recorded in the database with its start and end time.

Tip: start a scenario, then switch to *NETWORK > Topology & Inventory, FCAPS > Alarms* or *FCAPS > Performance*. Scenarios keep running while you move between pages.

34. Lab Simulator Checks

Some checks against a lab simulator can only be done on your network, for example an O-RAN-SC simulator in a Docker container. A practical order:

1. Confirm reachability of the **host port** (not necessarily 830), for example `Test-NetConnection 192.0.2.25 -Port 8830` in PowerShell. The SSH banner should appear with **TEST CONNECTION**.
2. Add the NF with that host port and the simulator's credentials. MiniSMO stores the password protected with Windows DPAPI.
3. **DISCOVER** and compare the capability matrix with what the simulator is expected to support.
4. **START MONITORING** and watch the *NETCONF Log* for `create-subscription / establish-subscription` replies and incoming `notification:*` entries.
5. Trigger events with the simulator's own tools (fault generation, configuration change from another client), and check that they appear in the Alarm Console, Configuration history and Performance.
6. For VES: check the VES step in the integration dialog after **SAVE**, then the **VES** card on the NF's **Monitoring** tab and *SYSTEM > Settings > VES Collector*. Watch heartbeat, fault and perf3gpp events arrive. If they don't, check that the simulator host can reach this PC on the collector port.

O-RAN-SC NTS devices typically sign in with keyboard-interactive, list their modules only in the YANG library, keep alarms in a 3GPP `AlarmList` in the running configuration, expose cell PM counters in their own YANG module, and (in current images) accept the 3GPP `NtfSubscriptionControl` for their VES destination. MiniSMO discovers all of this from the device's YANG library and modules, with no NTS-specific knowledge; you only enter the address, port and credentials.

A container port bound to `127.0.0.1` on the simulator host cannot be reached from another PC. MiniSMO does not change firewall rules on the simulator host or expose containers. The only change it makes to a simulator's configuration is its VES subscription entry, written when the NF is added (unless turned off).

35. Topology View

NETWORK > Topology & Inventory > Topology draws the network in three layers, top to bottom:

1. **O-CU**: O-CU-CP and O-CU-UP
2. **O-DU**
3. **O-RU**, grouped under the O-DU that serves them

External NFs appear in their layer by type, marked EXTERNAL. MiniSMO cannot read how external NFs are connected (there is no standard model it can rely on), so it lets you place them:

- An external **O-RU** starts in the group **not under an O-DU**. Select **PLACE EXTERNAL NF...**, or right-click the node and choose **Place in topology...**, and tick the O-DU(s) it is connected to over fronthaul.
- An external **O-DU** can be placed under the O-CU-CP (F1-C) and O-CU-UP (F1-U) that serve it.
- Clear every box to take the NF out of the hierarchy again.

Placement is stored in MiniSMO only and is never written to the NF; it is recorded in the audit log. The Demo Network topology is fixed. NFs of other types (GENERIC NETCONF NF) are listed under **OTHER NFs**.

The **Inventory** tab next to *Topology* shows the same NFs as a tree by region and site, with their cells. See [Inventory and Network Tree](#). Software versions are on *TOOLS > Software Campaigns*.

Each node shows:

Item	How it is shown
Name and type	Text on the node
Mode	SIM (cyan) or EXT (amber) badge
Connection	Green dot for CONNECTED, grey for DISCONNECTED
Operational state	Node dimmed when DISABLED
Alarm state	Labelled severity chip and border colour for the worst active alarm: red CRITICAL, orange MAJOR, yellow MINOR, blue WARNING

Select a node to open that NF in Network Functions. Right-click a node for **Open in Network Functions** and **Place in topology...** The view refreshes every 5 seconds, so you can watch scenarios play out.

See [Understanding the Demo Topology](#) for how the Demo Network is wired.

36. Inventory, Search and Notifications

36.1 Inventory and Network Tree

NETWORK > Topology & Inventory > Inventory (Community) shows the network as a tree: **Region** → **Site** → **NF hierarchy** → **Cells**. Parent and child NFs come from the topology (for example O-DU → O-RU), and cells from each NF's running configuration.

Region and site are inventory fields you set per NF: select an NF in the tree and choose **SET SITE / REGION...**, or use **Site / region** in the *Network Functions* toolbar. They are stored in MiniSMO only and never written to the NF. NFs without a region or site are grouped under *(no region)* and *(no site)*. An NF without its own site and region is shown under its topology parent (an O-RU under its O-DU). An NF whose site or region differs from its parent's is shown at its own site, and its node names the topology parent. An NF with the same site and region as its parent stays under the parent.

Each NF node shows its type, connection state and active alarms with the worst severity. Select a node to see its details. NF nodes have **OPEN NF**, **SET SITE / REGION...** and **INVESTIGATE** (marked **PRO**); cells have **INVESTIGATE CELL** (marked **PRO**). See [Investigate](#).

EXPORT CSV... asks for confirmation, then writes the inventory (name, type, address, vendor, model, serial number, software and hardware version, site, region, states, number of cells) to the exports folder and shows **Open file / Show in folder**. Software versions across the NFs are on [Software Campaigns](#).

36.2 NF Identity and Cells

MiniSMO reads each external NF's vendor, software version, NF type, model, serial number and hardware version (Community) with read-only `<get>` requests. It does this automatically when the NF connects, when it sends a VES `pnfRegistration`, and for any NF not read yet; **Discover identity** in the *Network Functions* toolbar reads it again on request.

Sources, in order. A later source only fills fields the earlier ones left empty:

Source	Fields
3GPP ManagedElement/attributes (TS 28.622), read with <code><get></code> and an explicit subtree filter; the standard O1 location	<code>vendorName</code> , <code>swVersion</code> , <code>managedElementTypeList</code> (NF type), <code>userLabel</code> , <code>locationName</code> , <code>managedBy</code>
<code>o-ran-software-management</code> , if the NF advertises it	build version and vendor code of the active, running software slot

Source	Fields
ietf-hardware (RFC 8348), if the NF advertises it	manufacturer, model, serial number, hardware and software revision
VES pnfRegistration (fallback)	vendor, model, serial number, software version, firmware version
VES event header (fallback)	nfVendorName (vendor), nfNamingCode (NF type code)

The `ManagedElement` leaves are operational (`config false`), so they are only in `<get>` replies, never in `<get-config>` , and MiniSMO never tries to write them. Some devices, including the O-RAN-SC NTS simulators, do not fill them in; the VES values are then used. `locationName` becomes the NF's site if you have not set one.

The dialog shows which source each value came from. A field is shown as unknown only when neither NETCONF nor VES reports it; nothing is guessed. Cells (NRCellDU, NRCellCU, EUTRANCell) are read from the latest configuration snapshot with their DN, administrative state, PCI and ARFCN.

Capabilities in the *Network Functions* toolbar summarises, vendor-neutral, what the NF can technically do (the **Capabilities** tab of the NF lists the raw NETCONF capabilities instead): configuration (writable datastore), candidate datastore, confirmed commit, rollback, validate, notifications, YANG-Push, FM, PM, software management, cell discovery and hardware inventory. It is derived from the NF's `<hello>` , YANG library and stream list.

36.3 Search

The search box at the top of the navigation pane (Community) finds NFs by name, address, vendor, model, serial number, software version, site or region; alarms by title, alarm ID, managed object or text; and cells by ID, DN or PCI. Select a result to open the NF or the alarm console.

36.4 Notifications

The **bell** at the top right of the window (Community) lists local notifications: automation alerts, approvals waiting for a decision, configuration drift, failed jobs, paused webhooks, monitoring method changes and fallbacks, storage state and license changes. A badge on the bell shows unread notifications; opening the list marks them read. Notifications stay on this PC. To send them elsewhere, use [webhooks](#) (Pro).

37. YANG Model Registry

SYSTEM > *Settings* > *YANG Models* is a local library of YANG modules. Use it to keep the models your devices implement next to the devices themselves, so you can read the schema while you look at `<get>` and `<get-config>` output.

The list shows, for each imported file:

Column	Meaning
Module	Module (or submodule) name
Revision	Latest <code>revision</code> date in the file
Namespace	XML namespace of the module
Prefix	Module prefix
Kind	module or submodule
Size / SHA-256	File size and hash, to check you have the exact file you expect

Select a model to read its source. **DELETE** removes it from the registry and from disk after confirmation.

Files are stored in `%LOCALAPPDATA%\Kaldrivon\MiniSMO\yang` as `<module>@<revision>.yang`.

MiniSMO does not ship third-party YANG models. Standards bodies and vendors publish YANG modules under their own terms; download them from the source you are entitled to use. The NF's Capabilities tab tells you exactly which modules and revisions a device implements.

MiniSMO does not generate screens from these models. When you edit an external NF's configuration, it uses the NF's own modules (read with `get-schema`) or the models imported here for names, types and constraints. See [Changing External Configuration](#) and [Known Limitations](#).

38. Importing a YANG Model

1. Open *SYSTEM > Settings > YANG Models*.
2. Select **IMPORT** and choose one or more `.yang` files.
3. MiniSMO checks each file and adds it to the list.

Checks performed on import:

- The file name ends in `.yang`, is UTF-8 text and is at most 5 MB.
- It contains a `module` or `submodule` statement with a valid name.
- Braces are balanced (comments and quoted strings are ignored), which catches truncated files.
- A `module` has a `namespace` statement.
- The same module and revision is not already imported.

These are sanity checks, not full YANG compilation. A file that passes can still have errors that a YANG compiler such as `pyang` would report.

Common import errors:

Message	Fix
only .yang files can be imported	Rename or pick the right file
unbalanced braces	The file is truncated or corrupted; download it again
module has no namespace statement	Not a valid YANG module
... is already imported	Delete the existing entry first if you want to replace it

39. Community and Pro

39.1 Community and Pro

Kaldrivon Mini SMO comes in two editions that share one installation, one database and one set of NFs.

Kaldrivon Mini SMO Community needs no license. It manages up to **5 external NFs** and includes every essential operation: onboarding, NETCONF and VES, capability and YANG discovery, inventory and topology, fault management with local acknowledgement and history, performance management, configuration management with confirmed changes, manual configuration backups and comparison, search, CSV exports, the network health report, system health, the audit log, and the simulated Demo Network. Simulated NFs never count against the limit.

Kaldrivon Mini SMO Pro is any valid signed license. It adds scale and the features that save time in day-to-day operations:

Area	Pro adds
Scale	The licensed number of external NFs instead of 5
Integration	Northbound REST API with tokens, O-RAN R1 for rApps (SME and DME), webhooks with routing, signing and retry
Automation	Rules , schedules and approvals for network changes
Configuration	Templates , desired state and drift , rollback , bulk operations , scheduled backups, comparison across NFs
Analytics	KPI rankings , biggest changes , alarm correlation , flapping
Troubleshooting	Investigate and diagnostic collection
Reports	Operations reports and scheduled reports
Software	Upgrade campaign planning
System	Retention for audit, backups and automation history; audit export

Pro features stay visible in Community and carry a small **PRO** marker. Ordinary FM, PM and CM work is never interrupted by upgrade messages.

39.2 Pro Features Without a License

Community includes a small allowance of every Pro feature, so each one can be tried on real NFs. Without a Pro license:

- each Pro page (Automation, Analytics, Software Campaigns, API & Webhooks, Configuration Operations, Reports, Audit Log) shows what Community includes of its features, and today's use, under the page title, with links to **Community vs Pro** and **License**;

- every Pro action (a button or a menu action) shows the same information in a notice at the top of the window before it runs;
- where the page note already shows the feature, no second notice is shown;
- actions that run something ask for confirmation, and without Pro the confirmation shows the allowance and today's use;
- going past the allowance stops the action, and the notice says which limit was reached;
- a Pro-only action (for example flapping suppression) is refused with a notice at the top of the window.

With a Pro license none of these notes appear. *SYSTEM* > *License* lists the whole allowance and today's use.

Pro feature	Community includes
Northbound REST API	1 API token, READ scope, 1,000 requests a day (listener on this PC only)
Webhooks	1 webhook, 200 deliveries a day (further deliveries wait for the next day)
Automation	2 rules, 1 of them scheduled; approvals for those rules
Alarm correlation	10 analyses a day, run on request (RUN ALARM CORRELATION)
Flapping	the flapping list and threshold (suppression needs Pro)
KPI rankings, most alarms and biggest changes	10 analyses a day, run on request (RUN KPI ANALYSIS). Counted separately from alarm correlation
Comparison across NFs or with a template	5 a day
Desired state and drift	1 desired state
Configuration templates	2 templates
Bulk operations	jobs on up to 3 NFs
Scheduled backups / scheduled reports	1 schedule each
Rollback	2 a day
Software campaign plans	2 a day
Operations reports	3 a day
Investigate	10 a day
Diagnostic collection	3 a day
Audit	search, 3 exports a day

Pro only: external NFs above 5; retention for audit, backups and automation history (the data retention presets for PM, FM and logs under *SYSTEM* > *Settings* > *Storage*

& *Retention* are Community); listening for the northbound API beyond this PC; flapping suppression; automation changes without approval. Daily limits reset at midnight, this PC's local time.

Items created with Pro beyond the allowance (rules, schedules, webhooks, tokens, templates, desired states) are kept after a downgrade. The oldest ones within the allowance keep running; the rest are paused, never deleted, and resume when a license is installed.

Licensing and NF capability are separate. A Pro license decides whether you may use a feature; the NF decides whether it can technically do it. Rollback, for example, needs both a Pro license (or the Community allowance) and an NF with a writable datastore. Past the allowance you see what Community includes; with Pro and an NF that cannot do it you see *not supported by this NF* and the reason.

Upgrade and downgrade keep your data. Installing a Pro license unlocks Pro features on the running installation: no reinstall, no re-onboarding, and alarms, PM history, configuration and inventory stay as they are. If a license expires or is removed, Pro actions become unavailable, stored data stays readable, and nothing is deleted or disabled. If more than 5 external NFs are onboarded, they keep being monitored; only adding more is blocked until the count is back within the allowance or a license is installed.

See [Installing a License](#).

40. Configuration Operations

40.1 Configuration Backups

FCAPS > Configuration > Configuration Operations > Backups (Community). **BACK UP NOW...** reads the NF's running configuration with a read-only `get-config` and stores it with a timestamp, the NF, its software version, your comment and who took it. Secret values are masked before storage, the same way as in configuration snapshots.

Backups are never overwritten: each one is a new entry. MiniSMO keeps the 50 newest backups per NF; with Pro you can protect individual backups (**KEEP**) and change how many are kept in *SYSTEM > Settings > General*, section **Pro operations**.

Backups taken automatically are marked by trigger: *PRE_CHANGE* before a template, approval or automation change, *PRE_ROLLBACK* before a rollback, *SCHEDULED* by a schedule, *BULK* by a bulk backup.

40.2 Comparing Configurations

Select a backup and choose **COMPARE WITH CURRENT** (reads the NF now) or **COMPARE WITH...** another backup. The result lists changed parameters with both values, and parameters found only on one side.

Comparison	Edition
Backup vs the NF's current configuration	Community
Backup vs another backup of the same NF	Community
Backup vs another NF, or ignoring list-key values (structure across NFs)	Pro
Backup or NF vs a template	Pro

Masked secret values (`*****`) are compared as masked.

40.3 Configuration Templates

PRO. *Configuration Operations > Templates* holds named sets of parameter values, for example a cell template:

```
NRCelldU[id=]/attributes/ssbPeriodicity = 20
NRCelldU[id=]/attributes/arfcnDL = 632628
```

Paths use the parameter IDs shown on the Configuration page. `*` matches any list-key value or one path segment; `**` matches several segments. A template can be limited to an NF type, a vendor and required YANG modules; MiniSMO refuses to apply it to an NF that does not match (*INCOMPATIBLE*).

PREVIEW ON NF... shows the changes for one NF without writing anything. **APPLY TO NFs...** is a [bulk operation](#).

40.4 Desired State and Drift

PRO. A desired state is a set of expected values (from a template or entered directly) for selected NFs. MiniSMO compares it with each NF's latest configuration snapshot every 15 minutes and on **EVALUATE NOW**:

State	Meaning
COMPLIANT	Every expected value matches
NON-COMPLIANT	At least one value differs
UNKNOWN	No snapshot yet, the parameter is not present, or the value is masked
NOT EVALUATED	Not checked yet

When an NF becomes non-compliant, MiniSMO raises a notification and a **DRIFT_DETECTED** event (for webhooks and automation). Nothing is corrected automatically. **PROPOSE REMEDIATION...** creates an [approval](#) with the changes that would restore the desired values.

40.5 Rollback

PRO. Select a backup and choose **ROLLBACK...**. MiniSMO reads the NF's editable parameters and lists every value that differs from the backup. Confirming runs:

1. **Validate**: the change is previewed against the NF's current values and YANG constraints.
2. **Backup**: the current configuration is saved (*PRE_ROLLBACK*).
3. **Apply**: candidate + validate + confirmed commit, or writable-running with test-then-set, as for any configuration change.
4. **Verify**: the values are read back before the commit is confirmed.

Rollback is available only when the NF has a writable datastore and configuration changes are enabled for it. With confirmed commit the NF reverts on its own if the change is not confirmed; without it, MiniSMO says so before you confirm. Rollback restores leaf values of entries that exist now; list entries added or removed since the backup are listed but not recreated or deleted. A rollback is never guaranteed to restore every value.

40.6 Bulk Operations

PRO. Bulk operations run one action on several NFs: configuration backup, read-only health check, or applying a template. Each shows the target list first; template pushes also show every change per NF and need confirmation. The job runs each NF

separately, so one NF failing does not stop the others, and shows a per-NF result. Jobs are listed on the *Jobs* tab.

41. Automation

41.1 Automation

PRO. *TOOLS* > *Automation* runs rules of the form **trigger** → **condition** → **action**.

Triggers	Conditions	Actions
Alarm raised or cleared	NFs, NF types	Local notification (routable by webhooks)
NF connected or disconnected	Minimum severity	Send the event to a webhook
Configuration changed	Alarm text contains	Health check (read-only)
Drift detected		Configuration backup
KPI threshold held for N minutes		Collect diagnostics
Schedule (interval or daily)		Health report
		Propose a configuration change (approval)

Example: *PRB utilisation DL > 90 for 30 minutes* → *notify, send webhook, run a health check*. A KPI rule fires once when the condition has held for the whole period and again only after it has cleared.

TEST (DRY RUN) describes what each action would do without doing it. Every run is recorded in *Run history* with its trigger, per-action results and failure reasons, and in the audit log. Storm limits: a cooldown per rule (default 15 minutes) and at most 30 runs per rule per hour.

41.2 Approvals

PRO. Actions that would change an NF never run on their own. They create an approval: **detect** → **analyse** → **recommend** → **a person approves** → **execute** → **verify** → **record**. The *Approvals* tab shows the target NF, each parameter with its current and proposed value, and who proposed it. **APPROVE AND APPLY** takes a backup, lets the NF validate the change, applies it with confirmed commit or test-then-set, reads it back, and records the result. **REJECT** records the decision; nothing is sent.

A rule can apply changes without approval only if two things are true: the rule's action has *Apply without approval* ticked, and an administrator turned on *Allow automation to apply changes without approval* in *SYSTEM* > *Settings* > *General*, section **Pro operations**. Both are off by default. Pending approvals expire after 24 hours (configurable).

41.3 Schedules

PRO. Schedules run configuration backups, health reports, operations reports and drift checks daily, weekly or every N minutes (at least 15), in this PC's local time. A schedule that was due while MiniSMO was closed runs once at the next start.

Schedules are created from the Backups tab, the Reports tab or *Automation > Schedules*, where they are listed with their next run and last result. **RUN NOW** asks for confirmation and runs a schedule once; jobs that change the network still create an approval first.

42. Analytics and Troubleshooting

42.1 Analytics

TOOLS > *Analytics* (marked **PRO**). With Pro the page refreshes every 30 seconds.

Without Pro nothing runs on its own: **RUN KPI ANALYSIS** (KPI rankings, most alarms, biggest changes) and **RUN ALARM CORRELATION** each ask for confirmation and use one analysis of their own daily allowance (see [Pro Features Without a License](#)). **Window (minutes)** sets the KPI window: 15, 60, 240 or 1440.

- **KPI rankings:** per KPI family (availability, PRB utilisation, throughput, connected users, RRC and handover success, drop rate, latency, packet loss, CPU), the NFs and resources from the worst end, over a chosen window. Families are recognised from the metric names the NFs report (3GPP TS 28.552, O-RAN and simulator names). A family without data shows *no data reported*; no KPI is derived or invented. Counters are ranked by rate.
- **Most alarms** in the last 24 hours.
- **Biggest changes:** the last hour against the 23 hours before it.
- **Alarm correlation:** active alarms raised close together (default 120 s) on topologically connected NFs or on the same managed-object subtree. Each group names a *potential parent alarm* and *likely related alarms*. This is a heuristic, not a root-cause claim.
- **Alarms counted:** simulated, external and NF supervision alarms (Loss of supervision, VES heartbeat missed). Health alarms raised on MiniSMO itself are left out.
- **Flapping:** alarms raised repeatedly (default 5 or more per hour). The list is shown in Community too. With *Do not notify or route flapping alarms* on, those raises produce no events for webhooks and automation; the alarms themselves are still shown.

42.2 Investigate

PRO. INVESTIGATE is available for an alarm (Alarm Console), an NF (**Investigate (Pro)** in the *Network Functions* toolbar) and an NF or cell in the network tree (**INVESTIGATE, INVESTIGATE CELL**). It collects what MiniSMO has stored about the target and lays it out as:

- **Observed data:** connection state and last error, active alarms and the last 24 hours of alarm history, configuration changes, monitoring gaps, actions taken from MiniSMO, KPI trends, alarms on connected NFs
- **Correlation:** facts that happened close together, for example a configuration change in the 30 minutes before an alarm
- **Possible explanation:** hedged readings ("may", "is possible"), never a diagnosis
- **Recommended check:** what to look at next

Data MiniSMO does not have (for example NF-internal logs) is not shown or guessed.

42.3 Diagnostic Collection

PRO. Diagnostics (Pro) in the *Network Functions* toolbar asks for confirmation, then writes one ZIP per NF to the exports folder and shows **Open file / Show in folder**. The ZIP contains: inventory entry (without credentials), capabilities and capability model, monitoring status, alarms and FM events, the latest configuration snapshot, configuration changes, monitoring gaps and the last 200 NETCONF messages. Configuration and NETCONF payloads are the redacted copies MiniSMO stores. It is read-only.

43. Reports and System Health

SYSTEM > Health & Capacity > Reports:

- **HEALTH REPORT** (Community): asks for confirmation, then creates the report and opens it. Contents: NFs connected, sites, cells, active alarms by severity, alarms raised and cleared in 24 hours, configuration changes, and a list of **Attention** and **Investigate** items (disconnected NFs, critical and major alarms, locked cells, non-compliant NFs, failed jobs, storage state), plus a table of all NFs. HTML and JSON.
- **OPERATIONS REPORT...** (Pro): for a period, per NF alarms raised and cleared, critical alarms, average time to clear, configuration changes and backups, a KPI summary and automation activity. HTML, JSON and CSV.
- **SCHEDULE...** (Pro): daily or weekly reports.

Reports are saved to `%LOCALAPPDATA%\Kaldriven\MiniSMO\exports\reports` and never overwrite each other. Simulated NFs are marked.

The **System health** card shows **HEALTHY**, **DEGRADED** or **CRITICAL** from storage, CPU, memory, the notification queue, NETCONF sessions, the event bus, failed jobs and failed webhook deliveries in the last 24 hours, and the alarm rate. The Dashboard shows the same state next to *What needs attention*, and lists each check that is not healthy there (for example *Memory: 91 % used*). The matching MiniSMO alarm (for example *High memory usage*) is raised only after two readings in a row, one minute apart, so a short peak does not create an alarm; the state on the Dashboard follows the current reading.

44. Integrations

44.1 Northbound REST API

PRO. *TOOLS > API & Webhooks > API access*. A separate, token-authenticated listener for OSS, ticketing systems and scripts. MiniSMO's own local API is not exposed.

1. Turn on the listener in *SYSTEM > Settings > General*, section **Pro operations** (address, port, HTTPS with the collector's self-signed certificate). The default is `127.0.0.1:8770`, this PC only.
2. **NEW TOKEN...**: a name and a scope. The token is shown once; only its hash is stored.
3. Call the API with `Authorization: Bearer <token>`. The OpenAPI description is at `/api/v1/openapi.json`, interactive docs at `/api/v1/docs`.

Endpoint	Scope
GET <code>/api/v1/nfs</code> , <code>/nfs/{id}</code> , <code>/inventory</code> , <code>/sites</code> , <code>/cells</code> , <code>/topology</code>	READ
GET <code>/api/v1/alarms</code> , <code>/kpis</code> , <code>/config/{nf}</code> , <code>/config/{nf}/backups</code> , <code>/health</code>	READ
POST <code>/api/v1/alarms/{id}/acknowledge</code> (local acknowledgement, never sent to the NF)	OPERATE
POST <code>/api/v1/nfs/{id}/backup</code> , <code>/nfs/{id}/health-check</code>	OPERATE
POST <code>/api/v1/automation/rules/{id}/run</code> (network changes still become approvals)	OPERATE

Errors are JSON `{"error": {"code", "message"}}` with a matching HTTP status: 401 for a missing or revoked token, 403 for scope or edition. Without a Pro license the oldest READ token works within the [Community allowance](#); past the daily request limit requests answer `429 ALLOWANCE_EXCEEDED`. API actions are recorded in the audit log as `api:<token name>`.

44.2 O-RAN R1 Interface

PRO. The [northbound listener](#) also serves the O-RAN R1 interface, so rApps can use MiniSMO as their SMO / Non-RT RIC framework for discovering services and for data. It uses the same address, port, HTTPS setting and API tokens; a READ token is enough, because registering an rApp, publishing an API or creating a data job never changes the network. *TOOLS > API & Webhooks > API access* shows how many rApps are onboarded, how many data jobs and published APIs exist. The endpoints are listed in the OpenAPI description (`/api/v1/openapi.json`, sections *R1 SME (CAPIF)* and *R1 DME*). Errors are RFC 7807 problem details (`application/problem+json`).

Service Management and Exposure (SME)

SME follows the CAPIF core functions of 3GPP TS 29.222, with the standard request and response bodies:

Function	Endpoint
Onboard an rApp (API invoker)	POST /api-invoker-management/v1/onboardedInvokers (needs <code>onboardingInformation.apiInvokerPublicKey</code> and <code>notificationDestination</code>); PUT / DELETE .../{onboardingId}
Register an API provider	POST /api-provider-management/v1/registrations (needs <code>regSec</code> ; each function in <code>apiProvFuncs</code> has a role AEF, APF or AMF and <code>regInfo.apiProvPubKey</code>); PUT / DELETE .../{registrationId}
Publish a service API	POST /published-apis/v1/{apfId}/service-apis (a <code>ServiceAPIDescription</code> with <code>apiName</code> and <code>aeFProfiles</code>); GET / PUT / DELETE .../{serviceApiId}
Discover service APIs	GET /service-apis/v1/allServiceAPIs?api-invoker-id=... with the optional filters <code>api-name</code> , <code>api-version</code> , <code>comm-type</code> , <code>protocol</code> , <code>aeF-id</code> , <code>data-format</code> , <code>api-cat</code>

MiniSMO assigns the identifiers (`apiInvokerId` , `apiProvDomId` , `apiProvFuncId` , `apiId`) and returns a `Location` header. A public key or `regSec` that is already registered, or an API name an APF already published, is refused with 403. Publishing needs a registered APF, and every `aeFId` must be a registered AEF. Removing a provider also removes the APIs its APFs published.

MiniSMO publishes its own APIs, so an rApp discovers them right after onboarding (they are also in the `apiList` of the onboarding answer): **data-consumer** (the DME below) and **minismo-northbound** (the [REST API](#)), both on AEF `AEF_Minismo` at the address the rApp used.

Not provided: CAPIF security (TS 29.222 CAPIF_Security_API, OAuth tokens per API) and CAPIF events. Access control is the API token.

Data Management and Exposure (DME)

DME follows the O-RAN-SC Information Coordination Service data consumer API (`/data-consumer/v1` , `snake_case` bodies). MiniSMO is the producer of four information types:

Information type	Data	Job definition (all optional)
<code>fm-alarm-events</code>	Alarms raised and cleared, as MiniSMO stores them (simulated and external NFs)	<code>nf_ids</code> , <code>severities</code> (CRITICAL, MAJOR, MINOR, WARNING, INDETERMINATE), <code>include_cleared</code> (default true)

Information type	Data	Job definition (all optional)
cm-change-events	Configuration changes: NETCONF change notifications and changes made from MiniSMO (changed by, datastore, paths)	nf_ids
pm-measurements	PM samples as collected: NF, metric, resource, time, value (counters as a rate per second over their interval)	nf_ids, metrics (names or wildcard patterns), period_s (60–86400, default 900)
nf-topology	A snapshot of the NFs and links when the job is created, then each NF connection change	nf_ids

GET /data-consumer/v1/info-types lists them and GET .../info-types/{infoTypeId} returns the JSON Schema of the job definition (job_data_schema). Create or change a job with PUT /data-consumer/v1/info-jobs/{infoJobId} :

```
{
  "info_type_id": "fm-alarm-events",
  "job_owner": "my-rapp",
  "job_result_uri": "https://rapp.example:8443/alarms",
  "job_definition": {
    "severities": ["CRITICAL", "MAJOR"]
  }
}
```

The answer is 201 for a new job and 200 for a change; a job definition that does not match the schema gives 400, an unknown information type 404, and changing the type of an existing job 409. MiniSMO sends the data as an HTTP POST of a JSON array to job_result_uri : events as they happen, PM once per period_s with the samples collected since the previous delivery (at most 10,000 per delivery; no empty deliveries). GET .../info-jobs lists jobs (filters infoTypeId, owner), GET .../info-jobs/{infoJobId}/status returns ENABLED with producer MiniSMO, and DELETE removes a job (or all jobs of an owner).

PUT /data-consumer/v1/info-type-subscription/{subscriptionId} with owner and status_result_uri subscribes to information type availability: MiniSMO immediately posts each type with its schema and status REGISTERED .

Without Pro, or with the listener turned off, nothing is delivered and jobs report DISABLED ; they resume when the listener is back. Failed deliveries are counted per job and not retried. Simulated NFs are marked as such in the data (simulated in the topology), and only data MiniSMO really has is sent.

44.3 Webhooks

PRO. *TOOLS > API & Webhooks > Webhooks* sends events as JSON POST s:

Setting	Meaning
URL	<code>https://</code> recommended
Events	Any of ALARM_RAISED, ALARM_CLEARED, NF_CONNECTED, NF_DISCONNECTED, CONFIG_CHANGED, DRIFT_DETECTED, DRIFT_RESOLVED, JOB_FINISHED, AUTOMATION_ALERT, KPI_THRESHOLD (none = all)
Minimum severity	Alarm events at or above this severity
Secret	Optional. The body is signed with HMAC-SHA256: header <code>X-Kaldrivon-Signature: sha256=<hex></code> . Stored protected, never shown again

SEND TEST asks for confirmation, posts a test event and shows the result. Without Pro it counts as one of the day's deliveries. Failed deliveries are retried after 30 s, 2 min, 10 min, 30 min and 1 h, then marked FAILED. To prevent storms, a webhook keeps at most 500 pending deliveries, at most 20 are sent per 5-second cycle, and a webhook pauses after 20 failures in a row (with a notification); enable it again after fixing the endpoint. Delivery status and the last error are listed below the webhooks.

45. Software Campaigns and Audit

45.1 Software Upgrade Campaigns

TOOLS > Software Campaigns (marked **PRO**) lists, under **SOFTWARE VERSIONS**, the versions the NFs report per NF type and vendor, with the count and the NFs on each version. Viewing the versions needs no license. **FINDINGS** highlights:

- **Outliers:** a version used by a small share of NFs of the same type and vendor
- **Unknown versions:** NFs that report no software version
- **Mixed versions within a site**

Versions are what the NFs report. MiniSMO does not judge vendor lifecycle status.

PRO. PLAN CAMPAIGN... plans a campaign: a name, a target version and the target NFs. For each NF MiniSMO records the current version and a readiness state: *READY* (the NF exposes `o-ran-software-management`), *UNSUPPORTED_BY_NF*, *ALREADY_AT_VERSION* or *UNKNOWN_VERSION*. MiniSMO does not download, install or activate software in this release; use the plan with the vendor's procedure.

45.2 Audit Log

LOG > Audit Log (Community) records who did what, when, to which target and with which result: NFs added, changed and removed, connections, monitoring, configuration changes and permissions, backups and rollbacks, alarm acknowledgements and notes, template and bulk operations, approvals, automation runs, schedules, webhooks and API tokens, settings and license changes. Actions from automation appear as `automation:<rule>`, from the API as `api:<token>`, and from the application as the Windows user.

Request bodies are never stored, so passwords, keys and tokens cannot reach the log. Community keeps a year of audit history, searches by text and action, and includes 3 CSV exports a day (**EXPORT CSV** asks for confirmation and shows **Open file / Show in folder**). Pro adds filters by source, result and time, unlimited export, and configurable retention.

MiniSMO is a single-user desktop application: there are no separate logins or roles in this release. Protect access to the PC and the Windows account.

46. Community Edition and Licensing

Without any license file MiniSMO runs as **Kaldrivon Mini SMO Community**, which manages up to **5 external NFs** with every essential FM, PM and CM function. A valid license makes it **Kaldrivon Mini SMO Pro**: the licensed number of external NFs plus the Pro features without limits. Community includes a small allowance of each Pro feature; see [Pro features without a license](#).

How capacity is counted:

- Only **EXTERNAL** NFs count. Simulated Demo Network NFs never count against your license, however many there are.
- Connected and disconnected external NFs **both** count. An external NF uses a slot from the moment you save it.
- **Deleting** an external NF releases its slot immediately.

When you need more than 5 external NFs or the Pro features, Kaldrivon issues a license file (`.klicense`) with a capacity, for example 10, 25, 50, 100 or 250 external NFs, or unlimited. A license file also carries a customer name, a validity period and optionally a list of features. Any valid license unlocks Kaldrivon Mini SMO Pro.

License files are cryptographically signed by Kaldrivon and validated locally on your PC every time MiniSMO starts. No Internet connection or activation server is involved. If a license file is edited in any way, it stops being valid and MiniSMO runs as Community.

To see your current status, open *SYSTEM > License* or look at the license indicator in the title bar. Without a Pro license, the License page also lists **Pro features in Community**: what Community includes of each Pro feature and today's use. Daily limits reset at midnight, this PC's local time.

Related: [Installing a License](#), [External NF Capacity](#).

47. Installing a License

1. Save the `.klicense` file you received from Kaldrivon somewhere on your PC, for example `Acme-Telecom-Kaldrivon-MiniSMO.klicense`.
2. Open *SYSTEM > License*.
3. Select **INSTALL LICENSE** and choose the file.
4. MiniSMO checks the signature, the product, the validity dates and (if the license is tied to one computer) the machine. If everything checks out, the page shows **Edition: Kaldrivon Mini SMO Pro, License status: Active** and the new allowance. Your NFs, alarms, PM history and configuration stay as they are; nothing needs to be reinstalled or re-onboarded.

The license is copied to

`%LOCALAPPDATA%\Kaldrivon\MiniSMO\license\license.klicense`. You can delete the original file afterwards, but keep a copy somewhere safe.

Installing a new license replaces the previous one. A license that has already expired is refused at install time.

To go back to Community, select **REMOVE LICENSE** and confirm. Your NFs and data are not touched; Pro features become unavailable.

VIEW DETAILS shows every field of the installed license: license ID, customer, edition, license type, valid-from and expiry dates, features, whether it is machine-bound, and the signing key ID.

Machine-bound licenses

Some licenses are tied to a single computer. If yours is, Kaldrivon will ask for your machine fingerprint before issuing it; **VIEW DETAILS** on the License page shows it, with a **COPY** button. A machine-bound license installed on a different PC shows *MACHINE_MISMATCH* and does not raise capacity.

48. Understanding External NF Limits

48.1 External NF Capacity and License Limits

The license page shows three numbers:

Field	Meaning
Managed NF allowance	How many external NFs your effective license allows (5 for Community, or the licensed number, or Unlimited)
Managed NFs	How many external NFs exist right now, connected or not, for example 3 / 5
Over Capacity	How far usage is above capacity (only shown when it is)

When usage reaches capacity, **ADD EXTERNAL NF** reports *Community supports up to 5 external managed NFs (5/5 in use). Upgrade to Kaldrivon Mini SMO Pro to manage additional NFs.* With Pro it reports the licensed limit, for example *Your Kaldrivon Mini SMO Pro license allows 25 external NFs (25/25 in use).* and nothing is created. Existing NFs keep working. The check is done by the MiniSMO backend itself, not only by the screen, so it also applies to scripts that call the local API.

To add more external NFs, either delete ones you no longer need, or install a license with higher capacity.

Simulated NFs do not use capacity. You can create the Demo Network at any time.

48.2 Expired License and Downgrades

When a subscription or trial license expires

- The status changes to **LICENSE EXPIRED**.
- MiniSMO runs as Community: capacity reverts to 5 external NFs and Pro features fall back to the [Community allowance](#).
- Nothing is deleted. All external NFs, their settings, logs, backups, templates, rules and captured data stay available and readable, monitoring of existing NFs continues, and existing sessions are not dropped.
- If you have more than 5 external NFs, you cannot add new ones until you delete enough to get under 5 or install a renewed license.

When you install a license with a lower capacity (for example 10, while 20 external NFs exist), MiniSMO shows:

- Licensed Capacity: 10
- Current Usage: 20
- Over Capacity: 10

All 20 NFs remain. New external NFs are blocked until usage is below the licensed capacity.

A license that is not valid yet (its start date is in the future) shows **NOT_YET_VALID** and runs as Community until the start date.

49. System Capacity Assessment

49.1 Capacity & Storage Overview

SYSTEM > Health & Capacity > Capacity & Storage answers three separate questions. MiniSMO keeps them apart on purpose, because they limit different things:

Concept	Question it answers	Where it comes from
License capacity	How many external NFs am I entitled to configure?	Your license (Community: 5)
System capacity	How many NFs can this computer handle at acceptable performance?	The capacity test you run on this PC
Storage capacity	How much history can I keep before cleanup or disk limits are reached?	Retention settings and free disk space

Example: a 25-NF license on a PC whose capacity test suggests about 180 external NFs, with enough disk for 30 days of 5-minute PM history. None of these numbers changes the others. In particular, the capacity test never reduces your license.

The page has three tabs:

- **System Capacity:** computer information, current load, the latest capacity test and **RUN CAPACITY TEST**.
- **Storage:** space used by category, free disk, storage health, growth forecast, **CLEAN UP NOW, PURGE DATA...**, exports and backups.
- **History:** previous capacity tests, with view, delete and export.

Retention (how long each kind of history is kept, presets, and the retention impact estimator) is on *SYSTEM > Settings > Storage & Retention*.

The Dashboard's **LICENSE AND CAPACITY** card shows the Managed NFs, Technical capacity and Storage state. **HEALTH & CAPACITY** on that card opens this page.

All measurements stay on your computer. MiniSMO does not upload machine details, capacity results or storage data.

49.2 Running a Capacity Test

1. Open *SYSTEM > Health & Capacity > Capacity & Storage*, **System Capacity** tab.
2. Choose the profiles to test (all are selected by default) and the test length: **Quick** (5 s per stage), **Standard** (15 s) or **Thorough** (60 s).
3. Select **RUN CAPACITY TEST**. Progress shows the current profile, stage and live measurements.
4. Select **CANCEL** at any time. Partial results are kept and temporary data is removed.

How the test works:

- It ramps up step by step: 25, 50, 100, 250, 500, 750 and 1,000 NFs. It never starts at 1,000.
- Each profile runs against a **separate temporary MiniSMO instance** with its own database, created in the `temp` folder and deleted afterwards. Your NFs, data, alarms and license usage are not touched.
- Test NFs are simulated, so the test works in Community and never uses external NF license slots.
- Each stage is watched every second. The test **stops ramping** when conservative safety limits are reached: sustained CPU above 90%, low available RAM, API P95 latency above 1 second, slow database writes, PM scheduling falling behind, errors, or low disk space. It does not try to push your PC to its limit.

Workload profiles:

Profile	What it simulates
Idle Simulation	NFs exist and keep state; minimal background activity
Typical Lab	PM every 5 s, occasional alarms and configuration changes, dashboard and topology reads
PM Intensive	PM every second, frequent chart and history reads
High Activity	PM every second, many alarms, scenario activity, configuration changes and NETCONF logging
Estimated External NF Management	Real NETCONF/SSH sessions to a local test server, each polled regularly

The external profile is an **estimate**. The test server runs on the same PC, which makes the result conservative, and it does not represent any particular vendor's equipment.

What is measured at each stage: total, backend and frontend CPU; total, backend and frontend RAM; available RAM; API P50 and P95 latency; database write latency; PM and scenario scheduling delay; dropped PM updates; delayed events; event backlog; error rate; NETCONF operation latency (external profile); database growth rate; disk write rate; average PM row size; and NETCONF, FM and scenario log growth.

Only stage summaries are stored. Raw per-second telemetry is discarded after the test.

49.3 Understanding Technical Capacity

For each profile the results show two numbers:

- **Maximum Tested:** the largest stage that completed without reaching a safety limit.
- **Recommended:** the number MiniSMO suggests for everyday use. It is never the maximum tested value.

The recommendation is the smallest of three dimensions, so you can see what limits you:

Dimension	How it is derived
CPU/RAM	Scaled from measured CPU at the maximum stage to a 60% CPU target, and limited by RAM headroom
API/Database	Scaled from measured API P95 and database write latency to 250 ms and 100 ms targets
Storage policy	How many NFs your free disk (above the warning threshold) can hold at steady state under the current retention settings

In addition, no recommendation exceeds 80% of the maximum tested value.

The results also show CPU, RAM, P95 latency and storage write rate at the recommended point, and the stage table shows every measurement.

Results depend on the MiniSMO version, this computer, what else is running, and your retention settings. Run the test again after hardware or settings changes. See [Capacity vs Storage Constraints](#).

50. Storage Management

The **Storage** tab shows where MiniSMO's disk space goes:

Category	Contents
Database	The whole SQLite database, including its <code>-wal</code> and <code>-shm</code> files
PM Data	PM samples and aggregates of the Demo Network, PM read from external NFs (NETCONF, YANG-Push, VES measurements) and file-based PM (O-RAN/3GPP XML files)
FM History	Active and cleared alarms of all NFs and the alarm event history
NETCONF & VES Logs	Stored NETCONF messages, received VES events and recorded monitoring gaps
Configuration History	Configuration snapshots, change notifications, configuration backups of all NFs and the simulated configuration history
Operations & Audit	Audit log, notifications, automation runs, background jobs, webhook deliveries and approvals
Scenario & Capacity History	Scenario runs and capacity test summaries
Inventory & Other	What is left of the database: NFs, sites, capabilities, monitoring settings, templates, rules, schedules, settings, the YANG registry and the database's own overhead (indexes and free pages). Usually small; free pages are reused for new data
YANG Models	Imported <code>.yang</code> files
Exports	Files you exported
Application Logs	Backend and application log files
Backups	Database backups

Below the table: **Total data usage**, **Available disk space**, and the storage health state. PM, FM, logs, configuration, operations and scenario sizes are parts of the database total.

The tab updates by itself when files are added or deleted in the data folder, for example when you delete backups or exports in File Explorer: every few seconds MiniSMO compares a light fingerprint of the folders (names, sizes and times only) and measures storage again only when it changed. Nothing is measured on a timer otherwise.

Other actions: **OPEN EXPORT FOLDER**, **DELETE SELECTED EXPORT**, **CLEAN EXPORTS**, **CREATE BACKUP**, and **EXPORT STORAGE REPORT**.

51. Understanding MiniSMO Data Storage

All MiniSMO data lives under `%LOCALAPPDATA%\Kaldrivon\MiniSMO` (see [Data Storage Locations](#)). The largest part is usually PM history in the database.

PM history is kept in three tiers:

Tier	Resolution	Default retention
Raw	Every sample (1 to 30 s)	6 hours
5-minute aggregates	Min, max, average and sample count per counter per 5 minutes	7 days
Hourly aggregates	Same, per hour	30 days

Charts pick the tier automatically: raw data for recent periods, then 5-minute and hourly aggregates for longer ranges.

The database uses SQLite in WAL (write-ahead log) mode, which lets the backend write while pages read. The WAL file (`minismo.db-wal`) is checkpointed back into the database after cleanups and on clean shutdown, and it is included in the storage figures.

Freed database space is reused for new data. The database file shrinks gradually through incremental vacuum after cleanups; MiniSMO does not run a full VACUUM during normal use because that would block the database.

52. PM Data Retention

52.1 PM Retention

PM retention controls how long each PM tier is kept (defaults, from the MINIMAL preset: raw 6 hours, 5-minute aggregates 7 days, hourly aggregates 30 days). PM history older than the hourly retention is deleted unless you exported it or changed the retention.

Raw samples are always aggregated into the 5-minute tier **before** they are deleted, and 5-minute aggregates are rolled up into hourly aggregates before they expire. Longer-range charts keep working after raw data is gone; they just show 5-minute or hourly resolution with min/max bands.

Retention is applied automatically (see [Automatic Cleanup](#)) and when you select **CLEAN UP NOW**. PM inside its retention window is never deleted, even when disk space is low.

The three tiers apply to simulated PM. PM samples from external NFs (all sources, including measurement files) are stored as received and kept as long as the 5-minute aggregates (7 days by default). They are not aggregated.

52.2 PM Aggregation

For every NF and counter, each aggregate record stores:

Field	Meaning
bucket_start / bucket_end	The 5-minute or hourly period
min / max	Lowest and highest sample in the period
avg	Average of all raw samples in the period
sample_count	Number of raw samples
sum	Only for counters where a sum is meaningful. The simulated counters are rates, levels or percentages, so no sums are stored

Hourly averages are weighted by sample count, so they equal the average of all raw samples in the hour.

Aggregation runs every 5 minutes in the background. It is idempotent: re-running it for the same period updates the existing records instead of creating duplicates, so restarts and overlapping runs are safe.

52.3 Why PM Data Can Grow Quickly

PM volume multiplies quickly:

$$\text{NFs} \times \text{samples per hour} \times \text{hours kept}$$

For example, 250 NFs sampled every 5 seconds produce 180,000 raw rows per hour, or 4.3 million per day. At about 330 bytes per row that is roughly 1.4 GB per day of raw PM before aggregation.

Aggregates are much smaller: one row per counter per 5 minutes (288 per day) or per hour (24 per day), regardless of the sample rate.

That is why MiniSMO keeps raw data for a short period only and relies on aggregates for long-term history. The biggest levers are the **PM Update Interval**, the **raw retention**, and the number of NFs. Use the [retention impact estimator](#) to see the effect before you change anything.

53. FM History Retention

Alarm history is kept for 30 days by default.

- **Active alarms are never removed** because of age. They stay until they are cleared.
- **Cleared alarms** are deleted once their clear time is older than the FM retention.
- External monitoring history (alarm events, configuration change notifications, VES events and monitoring gaps) is also kept for the FM retention. Current external alarms and the latest configuration snapshots are kept.

Change FM retention in *SYSTEM > Settings > Storage & Retention*.

54. NETCONF History Retention

NETCONF messages are kept for 7 days by default. Choose 7, 14, 30, 90 days or a custom value. You can also set a **maximum size** for NETCONF history (for example 500 MB, 1 GB, 5 GB, or no limit). When either the age or the size limit is exceeded, the oldest messages are removed first.

Large replies are capped in history by the **maximum stored payload size** (default 1 MB per message). A larger message is stored with its metadata and the first part of the XML, and marked **PAYLOAD TRUNCATED FOR HISTORY**; the log shows its original size. This only affects the stored history. The reply shown on the Configuration page's NETCONF read tab after a GET or GET-CONFIG is always complete.

55. Configuration History

Every time a simulated NF's configuration is applied or reverted, MiniSMO records one history entry per changed field:

Field	Example
Timestamp	2026-09-26 10:42
NF	O-RU-003
Revision	7
Field	pci
Old value / New value	110 → 222
Source	USER, SCENARIO or SYSTEM

Select **HISTORY** on the Configuration page to view it. The last 50 revisions per NF are kept by default; older revisions are removed by retention. The current configuration is never deleted by retention.

56. Automatic Cleanup

When **Automatic cleanup** is on (the default), MiniSMO applies retention in the background without blocking the application:

Job	When	What
Storage check	Every minute	Disk space and storage health
PM aggregation	Every 5 minutes	Builds 5-minute and hourly aggregates
Light cleanup	Hourly	Expired raw PM and NETCONF messages
Deep cleanup	Daily	All categories, then database maintenance
Protective cleanup	When storage becomes CRITICAL	Removes data already eligible under your retention policy

Cleanup schedule options: *Hourly and daily* (default), *Daily only*, or *Manual* (only **CLEAN UP NOW**). Protective cleanup still runs when storage is critical, but it only removes data that is already past its retention.

Retention never deletes: the license, the NF inventory and stored credentials, current configuration, YANG models, exports (unless export cleanup is enabled), the running scenario, active alarms, or PM inside its retention window.

57. Manual Cleanup

CLEAN UP NOW on the Storage tab first shows what would be removed, for example:

Item	Estimate
Raw PM beyond retention	1.2 GB
Expired NETCONF messages	180 MB
Expired application log files	42 MB
Expired scenario run history	8 MB
Estimated reclaimed	1.43 GB

Select **CLEAN UP** to run it or **CANCEL** to close. Only data past its retention is removed. The estimate is approximate; SQLite reuses freed space inside the database first, and the file shrinks gradually.

PURGE DATA... on the Storage tab deletes *all* data of the groups you tick. Each group shows how many rows (or files) it would delete:

Group	Deletes
FM	Alarms and alarm event history of all NFs
PM	PM samples and aggregates, PM from external NFs and file-based PM records
CM	Configuration snapshots, change notifications and simulated configuration history
LOGS	NETCONF message log, VES event log and monitoring gaps
BACKUPS	Configuration backups of all NFs, including ones marked KEEP
OPS	Audit log, notifications, automation runs, background jobs, webhook deliveries and approvals
FILES	Files in the data folder: exports, temporary files and database backup copies

Type **CONFIRMED** to enable the button. NFs and their settings, monitoring and file-PM settings, topology, MiniSMO settings, the license, YANG models, templates, rules, schedules, webhooks and API tokens are kept. The purge is recorded in the audit log (after an **OPS** purge, that record is the first new entry). A file still open in another program is left in place. To remove everything, use [Reset MiniSMO](#).

Exports are never removed automatically unless you turn on *Export cleanup*. Use **DELETE SELECTED EXPORT** or **CLEAN EXPORTS** to remove them yourself.

Backups: **CREATE BACKUP** makes a consistent copy of the database using SQLite's online backup method, safe while MiniSMO is running. A *Configuration* backup contains inventory, settings, the YANG registry, scenarios and configuration history,

but leaves out PM, NETCONF messages and alarm history to keep it small. A *Full* backup contains everything. Automatic backups (made before database upgrades) are limited to the last 10.

58. Disk Space Warnings

MiniSMO checks free space on the drive holding its data every minute.

State	Default condition	What happens
HEALTHY	Enough free space	Nothing
WARNING	Less than 10 GB or less than 10% free, whichever is more conservative	Notification and Dashboard warning; cleanup is recommended
CRITICAL	Less than 5 GB or less than 5% free	Protective cleanup of data eligible under retention runs automatically
CLEANUP RUNNING	A cleanup is in progress	Shown until it finishes

The Dashboard's **LICENSE AND CAPACITY** card shows the storage state: *Healthy*, *Warning*, *Critical* or *Cleanup running*. **HEALTH & CAPACITY** opens Capacity & Storage.

Optionally set a **Maximum MiniSMO managed data size** (10, 25, 50, 100 GB or unlimited). Approaching it gives a warning; reaching it triggers cleanup of eligible data.

If storage stays critical after cleanup, no more data is eligible for deletion: shorten retention, free disk space elsewhere, or remove exports and backups you no longer need. MiniSMO does not delete protected data to make room.

59. Capacity vs Storage Constraints

59.1 License Capacity vs Technical Capacity

The **License vs System** panel puts the numbers side by side:

Field	Example
Licensed External NF Capacity	25
Current External NFs	8
Estimated Technical External NF Capacity	~180
Typical Simulated NF Capacity	~350

If your license allows more external NFs than the capacity test suggests for this PC, MiniSMO shows an information message: *Your license permits more external NFs than this computer's current benchmark suggests for high-activity workloads.* This does **not** invalidate or reduce your license. It means that at full license usage this PC may become slow, and a more powerful PC or lighter workload may be needed.

A license with lower capacity than the computer can handle is equally normal. The license is what you are entitled to; the capacity test is what this PC can do.

59.2 Capacity vs Storage Constraints

A computer can have plenty of CPU and memory but too little disk for the history you ask it to keep. The capacity test therefore reports the storage dimension separately and explains when it is the bottleneck. For example:

Dimension	NFs
CPU/RAM	500
API/Database	400
Storage policy	250
Overall recommended	250

With a note such as: *Storage-constrained: CPU/RAM supports ~500 NFs but current PM retention settings limit this to ~250 NFs with the available disk.*

To raise the storage-constrained number, shorten PM retention (see [Changing Retention Settings](#)), use a longer PM interval, free disk space, or run MiniSMO on a PC with more disk.

60. Estimating Storage Requirements

60.1 Storage Forecast

The **Forecast** panel on the Storage tab estimates:

- current growth per hour and per day (PM, NETCONF and FM),
- the steady-state size under your retention settings,
- how many days it would take to reach the low-disk warning threshold **if cleanup were disabled**.

PM estimates use the row sizes measured in your own database once enough data exists; before that, calibrated defaults are used. NETCONF and FM growth come from the last 24 hours and 7 days of activity.

The **Retention impact estimator** lets you try settings before applying them. Enter NF count, counters per NF, PM interval, raw retention and aggregate retention to see the approximate disk usage. For example: 250 NFs, 12 counters, 5 s, 24 h raw, 30 d 5-minute and 90 d hourly.

Forecasts are estimates. Actual usage depends on the number of NFs, counters, PM frequency, NETCONF payload sizes, alarm frequency, scenario activity, retention settings and SQLite overhead.

60.2 Changing Retention Settings

Open *SYSTEM > Settings > Storage & Retention* and pick a preset, or set values individually:

Setting	MINIMAL (default)	STANDARD	EXTENDED
Raw PM	6 h	24 h	72 h
5-minute PM aggregates	7 d	30 d	90 d
Hourly PM aggregates	30 d	90 d	365 d
FM (cleared alarms)	30 d	90 d	365 d
NETCONF messages	7 d	14 d	30 d

Changing any of these values yourself switches the preset to **CUSTOM**, unless the values match a named preset exactly. MiniSMO never changes your preset on its own. **APPLY PRESET** loads a preset; **SAVE RETENTION SETTINGS** saves the values.

Other settings: application log retention (14 days), configuration revisions per NF (50), scenario history (90 days or 500 runs), capacity test history (50), NETCONF maximum size and maximum stored payload, automatic cleanup and schedule, disk warning and critical thresholds, maximum managed data size, and export cleanup (off).

Shortening a retention period takes effect at the next cleanup, or immediately with **CLEAN UP NOW**. Lengthening it only affects data that has not been deleted yet.

61. Exporting Capacity Reports

EXPORT CAPACITY REPORT on the System Capacity or History tab saves the latest (or selected) capacity test as:

- **PDF**: generated by MiniSMO from the HTML report,
- **HTML**,
- **JSON** with every result,
- **CSV** with the detailed stage measurements.

Files are saved to the exports folder as `Kaldrivon-MiniSMO-Capacity-Report-YYYYMMDD-HHMM.<ext>`. The report contains system information, license information, profiles, recommended and maximum tested NF counts, CPU/RAM and storage measurements, and this disclaimer: *Results are estimates based on the tested MiniSMO version, current computer, workload profile, retention settings, and system conditions. Results are not a guarantee of performance with all vendor equipment or production traffic patterns.*

Reports never contain device credentials or NF configuration.

62. Exporting Storage Reports

EXPORT STORAGE REPORT on the Storage tab saves a PDF, HTML, JSON or CSV report with: current total usage, size per category, retention settings, free disk space, estimated growth, estimated steady state, current warnings, and what is currently eligible for cleanup.

Files are saved to the exports folder as `Kaldrivon-MiniSMO-Storage-Report-YYYYMMDD-HHMM.<ext>`. Like all exports, they are created only when you ask and are never sent anywhere by MiniSMO.

63. Settings

SYSTEM > *Settings* has four tabs: **General**, **Storage & Retention**, **VES Collector** and **YANG Models**. The **General** tab holds application preferences. Changes are saved when you select **SAVE**.

Setting	Default	Notes
Theme	Dark	Dark, Light or System
Backend Port (restart required)	8765	Local port of the MiniSMO backend, 1024 to 65535. Takes effect after restarting MiniSMO. Change only if 8765 is used by another program
Auto-start Backend	On	Start the bundled backend automatically when MiniSMO opens
Default NETCONF Port	830	Pre-filled in the Add External NF dialog
NETCONF Connect Timeout (seconds)	10	1 to 120. Applies to TCP, SSH, authentication and the hello exchange
NETCONF Operation Timeout (seconds)	30	1 to 600. Maximum wait for an RPC reply
PM Update Interval (seconds)	5	1, 5, 15 or 30 seconds between stored simulated PM samples
Simulated alarm activity	normal	off, low, normal or high: how often Demo Network NFs raise, escalate and clear alarms on their own (about 1, 3 or 12 per NF per hour). Also on <i>SIMULATION</i> > <i>Demo Network</i>
Logging Level	INFO	DEBUG, INFO, WARNING or ERROR for the backend log
Update Channel	Stable	Stable, Preview or Internal. Reserved for future automatic updates; 0.4.0 does not check for updates
Show recent Help topics	On	Show recently viewed topics in the Help Center

The **Integration** section decides what happens when you add an external NF:

Setting	Default	Notes
Connect and start monitoring new NFs	On	Connect, discover and start FM / CM / PM monitoring as soon as the NF is saved

Setting	Default	Notes
Point new NFs at the VES collector (and keep them pointed here)	On	Write the collector address into the NF's VES settings and keep it pointed there. See Pointing an NF at the Collector

The **Pro operations** section (marked **PRO**) holds the settings of the Pro features. Without Pro, the settings marked *Pro only* cannot be changed.

Setting	Default	Notes
Northbound REST API listener (token-authenticated)	Off	See Northbound REST API
Listen on address	127.0.0.1	This PC only. Another address is Pro only
Port	8770	1024 to 65535
HTTPS (self-signed)	On	Uses the collector's certificate
Flapping: raises per hour	5	2 to 100
Do not notify or route flapping alarms	Off	Pro only
Correlation window (s)	120	10 to 3600
Approvals expire after (h)	24	1 to 720
Allow automation to apply changes without approval (administrator)	Off	Pro only
Backups kept per NF	50	Pro only to change; protected backups are not counted
Audit log retention (days)	365	Pro only to change; 30 to 3650
Automation history (days)	90	Pro only to change; 7 to 3650

The **VES collector** section holds the listener settings: **Collector enabled** (on), **Listen on address** (0.0.0.0 , all network interfaces), **HTTPS port (self-signed)** (8443), **HTTP listener** (on) and **HTTP port** (8080), **Collector address NFs should use** (empty: the local address that routes to each NF), **Protocol NFs should use** (https), **Authentication NFs must use** (none, or basic with **Username** minismo and a password). See [VES Collector](#).

The **NETCONF Call Home** section (off by default) has **Listen for Call Home connections**, the listen address (0.0.0.0), the port (4334) and **Wait for a call (seconds)**, how long a connect waits for an NF to call (60). It shows the listener state, the calls accepted and refused, a firewall hint, and the **REFUSED CALLERS** with **ADD AS NF...** and **CLEAR REFUSED LIST**. See [NETCONF Call Home](#).

The **Storage & Retention** tab holds the retention policy (presets MINIMAL (default), STANDARD and EXTENDED, or CUSTOM), automatic cleanup and schedule, export

cleanup, disk thresholds, the maximum managed data size and the retention impact estimator. See [Changing Retention Settings](#).

Invalid values are refused with a message explaining the allowed range.

64. Logs

MiniSMO writes these logs:

Log	Location	Content
Backend log	%LOCALAPPDATA%\Kaldrivon\MiniSMO\logs\backend.log	Startup, migrations, errors, NETCONF session events. JSON lines, rotated at 10 MB, 10 files kept
Application log	%LOCALAPPDATA%\Kaldrivon\MiniSMO\logs\app.log	Windows application events, backend start attempts, UI errors. Rotated at 10 MB, 10 files kept
NETCONF Log	In the database, see <i>LOG > NETCONF Log</i>	Every NETCONF message with full XML

About > OPEN LOG FOLDER opens the logs folder in File Explorer.

Rotated log files older than the application log retention (14 days by default) are deleted by cleanup; the current log file is never deleted. Passwords are never written to any log. If you send logs to support, review them first: they can contain host names, IP addresses and device configuration.

65. Data Storage Locations

MiniSMO keeps program files and your data separate.

What	Where
Program files	C:\Program Files\Kaldrивon\MiniSMO (or the folder chosen at install)
All user data	%LOCALAPPDATA%\Kaldrивon\MiniSMO
Database	...\MiniSMO\data\minismo.db (SQLite)
Logs	...\MiniSMO\logs
Imported YANG models	...\MiniSMO\yang
Exports (alarms, inventory, PM, NETCONF Log, audit, snapshots, diagnostics)	...\MiniSMO\exports
Reports	...\MiniSMO\exports\reports
Automatic database backups	...\MiniSMO\backups
Temporary files	...\MiniSMO\temp
Installed license	...\MiniSMO\license\license.klicense

%LOCALAPPDATA% is usually C:\Users\<you>\AppData\Local. Type it into the File Explorer address bar to open it.

Data is per Windows user. Another user on the same PC has a separate inventory and license.

66. Backup Considerations

What to back up:

- `%LOCALAPPDATA%\Kaldrivon\MiniSMO\data\minismo.db` : inventory, configuration, alarms, PM history, NETCONF Log, settings.
- `%LOCALAPPDATA%\Kaldrivon\MiniSMO\yang` : imported YANG files.
- Your original `.klicense` file.

Close MiniSMO before copying the database, so the backend is not writing to it.

Stored device passwords are encrypted with Windows DPAPI for your Windows account on this PC. If you restore the database on another PC or under another Windows user, the passwords cannot be decrypted; edit each external NF and enter its password again.

MiniSMO makes an automatic copy of the database in `backups` before applying any database migration during an upgrade. To make a backup while MiniSMO is running, use **CREATE BACKUP** on *SYSTEM > Health & Capacity > Capacity & Storage > Storage*: it uses SQLite's online backup method, so the copy is consistent. A *Configuration* backup leaves out large PM, NETCONF and alarm history; a *Full* backup includes everything. See [Storage Cleanup](#).

67. Software Updates

MiniSMO 0.4.0 does not check for updates automatically. To update, download the new `Kaldrivon-MiniSMO-<version>-Setup.exe` and run it over the existing installation:

- Your data folder is kept.
- The backend stops, program files are replaced, and on next start the database is backed up and migrated.
- Help content and the User Guide are replaced with the versions matching the new application.

The Help Center shows the help content version and the application version. If they ever differ, a warning is displayed.

68. Resetting and Uninstalling

68.1 Reset MiniSMO

RESET MINISMO... at the bottom of *SYSTEM > Settings* makes MiniSMO like a fresh installation. It deletes everything in the data folder: all NFs and their integrations, monitoring, the database with all PM, FM and CM history, VES and NETCONF logs, configuration backups, templates, rules, schedules, webhooks, API tokens, settings, stored credentials, YANG models, exports, application logs, certificates and caches.

You confirm three times, each time by typing a word:

1. **RESET**: the dialog lists what is deleted. Choose whether to **Keep the installed license** (on by default; when off, MiniSMO starts in Community edition) and whether to **de-integrate the external NFs first** (on by default).
2. **DELETE ALL DATA**
3. **MINISMO**: a progress window de-integrates each external NF (removes MiniSMO's own VES subscription, stops monitoring and closes the session), then schedules the reset. When you close it, MiniSMO restarts and the backend deletes everything *before* it opens the database or any other file.

Notes:

- De-integration removes only what MiniSMO itself wrote: its 3GPP `NtfSubscriptionControl` entry or, on an O-RU, its O-RAN WG4 configured subscription and heartbeat recipient. Other managers' entries are never touched. An NF that cannot be reached (or has no standard subscription model) is listed in the progress window; it keeps sending VES to this PC until you change it on the NF, and its events are stored as unattributed until you add the NF again. Without de-integration nothing is sent to any NF.
- The reset is recorded only in the reset result file; the audit log is deleted with everything else. Settings shows the time of the last reset, and how many items could not be deleted because another program had them open.
- Until MiniSMO restarts nothing is deleted, and **CANCEL PENDING RESET** keeps all data.
- In the dev build (the backend is started by Run-MiniSMO-Dev, not by the app), close MiniSMO and start Run-MiniSMO-Dev again after the third step; the reset runs when the backend starts.
- To keep only part of the data, use **PURGE DATA...** on *Capacity & Storage > Storage* instead (see [Storage Cleanup](#)).

68.2 Uninstalling

1. Close MiniSMO.
2. Open *Settings > Apps > Installed apps*, find **Kaldrivon MiniSMO** and select **Uninstall**. You can also use the Start Menu entry *Uninstall Kaldrivon MiniSMO*.

3. The uninstaller stops the backend and removes the program files and shortcuts.

Your data in `%LOCALAPPDATA%\Kaldrivon\MiniSMO` is **kept**, so reinstalling brings everything back. To remove it as well, delete that folder manually after uninstalling. This also removes your installed license, so keep a copy of the `.klicense` file.

69. Troubleshooting

69.1 Troubleshooting

Start with the symptom:

Symptom	Go to
MiniSMO opens but says the backend could not start	Backend Startup Problems
TEST CONNECTION or CONNECT fails	NETCONF Connection Troubleshooting
License not accepted, or capacity not what you expect	Licensing Troubleshooting
An error message you don't recognise	Common Error Messages
Scenario won't start	Create the Demo Network first
Configuration page has no edit buttons	The selected NF is external and changes are not enabled for it. Choose ENABLE CHANGES... (see Changing External Configuration)
No PM data	Simulated NFs: check the Demo Network exists. External NFs: check the NF's Monitoring tab (PM state, METHODS & FALLBACKS); see External Performance Data
Storage Warning or Critical on the Dashboard	Storage Warnings
PC slow with many NFs	Run a capacity test: Running a Capacity Test

When you contact support, include the MiniSMO version (**About**, at the bottom of the navigation pane), what you did, the exact error text, and the backend log.

69.2 Backend Startup Problems

The MiniSMO window talks to a local backend process, `KaldrivonMiniSMO.Backend.exe`, on `http://127.0.0.1:8765`. If it cannot reach the backend within about 30 seconds, it shows **Backend failed to start** with the reason.

Cause	What you see	What to do
Port 8765 is used by another program	<i>port 8765 is in use</i> or a health check from a different service	Change Backend Port in Settings (the diagnostic panel lets you pick a new port), then restart MiniSMO. Or run <code>netstat -ano</code> and look for port 8765 to find the program
Backend blocked by antivirus or application control	Backend exits immediately	Allow <code>C:\Program Files\Kaldrивon\MiniSMO\backend\KaldrивonMiniSMO.Backend.exe</code>
Damaged installation	<i>backend executable not found</i>	Reinstall MiniSMO (your data is kept)
Data folder not writable or disk full	Error in backend log about <code>minismo.db</code>	Free disk space; check permissions on <code>%LOCALAPPDATA%\Kaldrивon\MiniSMO</code>
Database damaged	Migration or database error	Close MiniSMO, move <code>data\minismo.db</code> elsewhere, start again (a new database is created), and restore a backup from <code>backups</code> if needed
Auto-start Backend is off	"Backend not running"	Select Start backend in the diagnostic panel, or turn the setting back on

The diagnostic panel has **Retry**, **Open log folder** and **Open startup troubleshooting** (this topic). The backend log is at `%LOCALAPPDATA%\Kaldrивon\MiniSMO\logs\backend.log`.

70. NETCONF Troubleshooting

Run **TEST CONNECTION** and look at the first stage that failed.

Problem	Possible causes	Actions
Host cannot be resolved	Wrong host name; DNS not available; name only resolvable on VPN	Check the spelling; try the IP address; check network and VPN connection
TCP 830 failed (timeout or refused)	Firewall between PC and device; wrong port; NETCONF disabled; device down	<code>Test-NetConnection <host> -Port 830</code> ; confirm the port on the device; check Windows Firewall and network ACLs; confirm the device is powered and reachable
SSH handshake failed	Port answers but is not SSH; incompatible SSH algorithms	Confirm the port; check the device supports modern SSH key exchange and ciphers
Host key changed	Device replaced or re-keyed; or someone intercepting the connection	Verify with the device owner. If legitimate, use Forget host key and connect again
SSH authentication failed	Wrong username or password; account locked; unsupported auth method (public key only)	Re-enter credentials with EDIT; check the account on the device; enable password or keyboard-interactive auth
NETCONF subsystem unavailable	SSH works but the <code>netconf</code> subsystem is not configured on this port or for this user	Enable NETCONF over SSH on the device; check you are using the NETCONF port, not the CLI port
Server Hello timeout	Incompatible NETCONF implementation; device NETCONF service hung or overloaded; timeout too short	Retry; raise NETCONF Connect Timeout ; restart the device's NETCONF service
Server hello is not well-formed XML	Non-compliant implementation	Capture the hello from the NETCONF Log and report it to the device vendor
Datastore not supported	Device did not advertise <code>:candidate</code> or <code>:startup</code>	Use <i>running</i>

Problem	Possible causes	Actions
RPC error in reply	Device refused the request (access control, bad filter)	Read the error message in the reply; check NACM permissions; remove or correct the subtree filter
Timed out waiting for server response	Large reply; slow device	Use a subtree filter; raise NETCONF Operation Timeout
Community supports up to 5 external managed NFs (or the licensed limit)	License capacity is used up	Delete an unused external NF or install a higher-capacity license

Every attempt, successful or not, leaves the exchanged XML in the NETCONF Log, which helps when you talk to the device vendor.

71. Licensing Troubleshooting

Problem	Likely cause	What to do
<i>License signature is invalid</i>	The file was edited, re-saved by an editor that changed it, or damaged in transfer	Install the original file you received. Do not open and save it in a text editor
<i>Not a Kaldrivon license file / license file is not valid JSON</i>	Wrong file, or an email client changed it	Ask Kaldrivon to resend it as an attachment or in a zip
<i>License signing key ... is not trusted by this build</i>	The license was issued with a newer signing key than this version knows about	Update MiniSMO, or ask Kaldrivon for a license issued for this version
<i>This license is not for Kaldrivon MiniSMO</i>	License for a different Kaldrivon product	Ask Kaldrivon for the MiniSMO license
Status LICENSE EXPIRED	The subscription or trial end date has passed	Renew. Your data is kept
Status NOT_YET_VALID	The license start date is in the future	Wait for the start date or check your PC's clock
Status MACHINE_MISMATCH	Machine-bound license installed on another PC, or Windows was reinstalled	Contact Kaldrivon support with your new machine fingerprint
<i>Community supports up to 5 external managed NFs ...</i>	Usage equals capacity	Delete unused external NFs or install a higher-capacity license
Status INVALID after it used to work	The stored license file was modified or damaged	Reinstall the original license file

If none of these apply, see [Support Information](#).

72. Common Error Messages

Message	Meaning	Where to look
Community supports up to 5 external managed NFs (5/5 in use)	No free external NF capacity	External NF Capacity
A network function named '...' already exists	Names must be unique	Pick another name
Password is required	Password and keyboard-interactive sign-in need a password	Add External NF dialog
A private key is required for public-key authentication	SSH key sign-in chosen without a key	SSH Authentication Methods
NETCONF port must be between 1 and 65535	Invalid port	Use 830 unless the device uses another port
Host Resolution: cannot resolve '...'	DNS failure	NETCONF Connection Troubleshooting
TCP Connection: timed out after 10s	Device unreachable or port filtered	NETCONF Connection Troubleshooting
Authentication: authentication failed	Wrong credentials	NETCONF Connection Troubleshooting
NETCONF subsystem: ... refused	NETCONF not enabled on that port	NETCONF Connection Troubleshooting
The device does not advertise the '...' datastore	Unsupported datastore	Running, Candidate and Startup Datastores
A parameter of an external NF cannot be edited	Changes are not enabled for that NF, or the parameter is a key, <code>config false</code> or not in a writable datastore	How MiniSMO Writes to External NFs
alarms can only be generated on simulated network functions	FM generation is simulator-only	About Simulated Alarms
No simulated network functions. Create the Demo Network first.	Scenarios need the Demo Network	Creating the Demo Network

Message	Meaning	Where to look
The Demo Network already exists	Only one Demo Network	Use RESET on <i>SIMULATION > Demo Network</i>
License signature is invalid	License file modified or damaged	Licensing Troubleshooting
LICENSE EXPIRED	Subscription or trial ended	Expired License and Downgrades
Backend failed to start	Local backend unavailable	Backend Startup Problems
This action is destructive; repeat the request with confirm=true	API call without confirmation (scripts only)	Add <code>confirm=true</code>

73. Security Notes

- **Local only API.** The MiniSMO backend API listens on `127.0.0.1` (this PC only). It is not reachable from the network. It has no user authentication, so do not expose it through port forwarding or proxies.
- **Northbound REST API (Pro).** Off by default. When you turn it on it listens on `127.0.0.1:8770` (HTTPS with the collector certificate) unless you choose another address, and every request needs a bearer token. Tokens are shown once and stored only as a SHA-256 hash; revoke a token as soon as it is no longer needed.
- **Webhooks (Pro)** send event data to the URLs you configure. Use HTTPS and a signing secret; the secret is encrypted with DPAPI and never shown again.
- **Audit log.** Changes made in MiniSMO (settings, licenses, NFs, configuration, automation, tokens) are recorded with the Windows user name. Passwords, secrets and tokens are never written to the audit log.
- **VES collector.** The VES event listener is the one network-facing part: by default HTTPS on 8443 and HTTP on 8080, on all interfaces, allowed through Windows Firewall on Domain and Private networks by the installer. It only accepts VES event POSTs. Turn on HTTP Basic authentication in Settings when NFs outside a closed lab can reach this PC, or switch the collector off.
- **Device passwords and SSH keys** are encrypted with Windows DPAPI under your Windows account before they are stored. They are never written in plain text to the database, logs or exports, and are never shown again after you enter them.
- **SSH host keys** are pinned on first connect. A changed key stops the connection until you confirm the change with **Forget host key**.
- **Writes to external devices are limited:** the VES destination (set automatically when the NF is added, unless turned off) and configuration changes you enable and confirm per NF. One more request acts on the NF without changing its configuration: a measurement file upload (`file-upload`) that you confirm. Every other write is refused before it is sent.
- **Licenses** are signed by Kaldrivon and checked locally on every start. Changing any value in a license file invalidates it. Licensing controls capacity and Pro features, never access to your data: removing or letting a license expire keeps all data, and Pro items are kept but paused.
- **Help and documentation** are bundled and rendered locally. The Help Center does not run scripts or load anything from the Internet.
- **Capacity and storage data** stay on your computer. Machine details, capacity results and storage figures are never uploaded; reports are only created when you export them.
- **Logs and exports** can contain device names, addresses and configuration. Treat them as you would any network documentation.
- **Updates:** only install MiniSMO from Kaldrivon and check the SHA-256 hash published with each release.

74. Known Limitations

Version 0.4.0 has these limitations:

- Configuration writes to external NFs happen only for NFs where you have enabled editing, after a preview and your confirmation (automation actions that change the network always wait for approval).
- No user accounts or role-based access control; one Windows user. Northbound API tokens have READ or OPERATE scope only.
- Software campaigns are planned and checked, not executed: MiniSMO does not download or activate software on NFs.
- Webhook delivery is at-least-once with retries; receivers should tolerate duplicates.
- Scheduled jobs run only while MiniSMO is running.
- VES provisioning uses the 3GPP `NtfSubscriptionControl` model (TS 28.623) and, for O-RUs, the O-RAN WG4 configured subscription (`o-ran-ves-subscribed-notifications`). For NFs with neither, set the VES destination with the vendor's tools; the collector accepts their events anyway. Neither model has credential fields, so collector credentials are set on the NF. Event-collector discovery by DHCP (O-RAN WG4) is not used.
- No complete YANG-driven user interface. YANG models are stored and displayed, and used for constraints when you edit an external NF's configuration.
- No configured (persistent) NETCONF subscriptions; subscriptions are dynamic and end with the session.
- No guaranteed O-RAN conformance; MiniSMO is not a conformance test tool.
- No A1, O2 or E2 interfaces.
- No Non-RT RIC and no rApp runtime. The R1 interface (Pro) offers CAPIF service discovery and data jobs; CAPIF security and CAPIF events are not provided, and the DME information types are MiniSMO's own.
- No high availability; single PC, single backend process.
- No multi-user access; one Windows user, local only.
- No Kubernetes or container platform required (or supported).
- Topology relationships for external NFs are not discovered; you place external O-RUs and O-DUs yourself (stored in MiniSMO only).
- One scenario can run at a time.
- Capacity test results are estimates for the tested PC and workload profiles, not guarantees for all vendor equipment or production traffic.
- External device compatibility depends on each device's NETCONF behaviour. Devices that deviate from RFC 6241/6242 may fail the hello exchange.

75. Glossary

Term	Meaning
Alarm	A notification that a fault condition exists on an NF, with a type and severity. Cleared when the condition goes away
Capability	A feature or YANG module a NETCONF server announces in its hello message, identified by a URI
CM	Configuration Management: reading and changing NF configuration
Datastore	A named copy of configuration on a NETCONF server: running, candidate or startup
External NF	A real network function managed by MiniSMO over NETCONF/SSH. Counts against license capacity
FM	Fault Management: alarm handling
NETCONF	Network Configuration Protocol (RFC 6241), XML-based protocol for managing network devices
NF	Network Function: a network element such as an O-RU or O-DU
O1	O-RAN interface between the SMO and O-RAN network functions for FCAPS management, based on NETCONF/YANG among others
O-CU-CP	O-RAN Central Unit, Control Plane: RRC and PDCP-C, controls one or more O-DUs over F1-C
O-CU-UP	O-RAN Central Unit, User Plane: PDCP-U/SDAP, carries user traffic to and from O-DUs over F1-U
O-DU	O-RAN Distributed Unit: real-time L2 and upper PHY, serves one or more O-RUs over fronthaul
O-RAN	Open RAN architecture specified by the O-RAN Alliance, with open interfaces between RAN components
O-RU	O-RAN Radio Unit: lower PHY and RF, connected to an O-DU over the open fronthaul
PM	Performance Management: counters and KPIs such as PRB utilization and throughput
RPC	Remote Procedure Call: a NETCONF request such as <code><get></code> , answered with an <code><rpc-reply></code>
Scenario	A scripted sequence of simulated events, such as a cell outage
Simulation	MiniSMO's built-in generation of synthetic NF state, alarms and counters
SMO	Service Management and Orchestration: the O-RAN management layer that hosts FCAPS functions and the Non-RT RIC
SSH	Secure Shell: the encrypted, authenticated transport NETCONF runs over

Term	Meaning
YANG	Data modeling language (RFC 7950) that defines the structure of NETCONF configuration and state data
Hello	The first NETCONF message each side sends, carrying capabilities and (from the server) the session ID
Session ID	Number the NETCONF server assigns to a session in its hello
Message ID	Identifier on each RPC, echoed in the matching reply
Subtree filter	XML fragment that limits what a <code><get></code> or <code><get-config></code> returns
Host key	The SSH server's public key; MiniSMO pins its fingerprint on first connect
.klicense	Kaldrivon license file format; signed, validated locally
PRB	Physical Resource Block: the unit of radio resource allocation in NR
PCI	Physical Cell Identity (0 – 1007 in NR)
NR-ARFCN	NR Absolute Radio Frequency Channel Number, identifies the carrier frequency
TAC	Tracking Area Code
MCC / MNC	Mobile Country Code / Mobile Network Code, together the PLMN identity

76. Support Information

76.1 Support Information

Before contacting support, collect:

1. The version from **About** at the bottom of the navigation pane (and the Help content version shown in the Help Center).
2. What you were doing and the exact error text (the **COPY** buttons help).
3. The backend log from `%LOCALAPPDATA%\Kaldrivon\MiniSMO\logs` (**About > OPEN LOG FOLDER**).
4. For NETCONF problems: the relevant NETCONF Log entries exported as JSON.
5. For licensing problems: your License ID (shown on *SYSTEM > License*). Do not send the license file unless asked.

Contact: info@kaldrivon.com Website: <https://kaldrivon.com>

76.2 About MiniSMO

About (at the bottom of the navigation pane) shows:

- Product name and version: **Kaldrivon MiniSMO 0.4.0**
- Description: RAN operations without the Tier-1 OSS footprint: O1 / NETCONF and VES management of O-RAN network functions, with a built-in simulated network for labs and demos.
- The data folder.
- Company: Kaldrivon, © 2026 Kaldrivon, <https://kaldrivon.com>

Buttons: **OPEN USER GUIDE** (opens the installed HTML User Guide), **OPEN HELP**, **VIEW LICENSE** and **OPEN LOG FOLDER**.

Kaldrivon MiniSMO is an independent product and is not represented as certified or endorsed by the O-RAN Alliance, 3GPP, ETSI, any network operator, or any equipment vendor.

Third-party open-source components and their licenses are listed in `THIRD-PARTY-NOTICES.txt` in the installation folder.

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